Spring 2025 - CIS 492/EEC 492/CIS 593/CIS 675/LAW 616
Assignment 2

Exercise 1: Scenario 2 - Spectre Attack Scenario Lab

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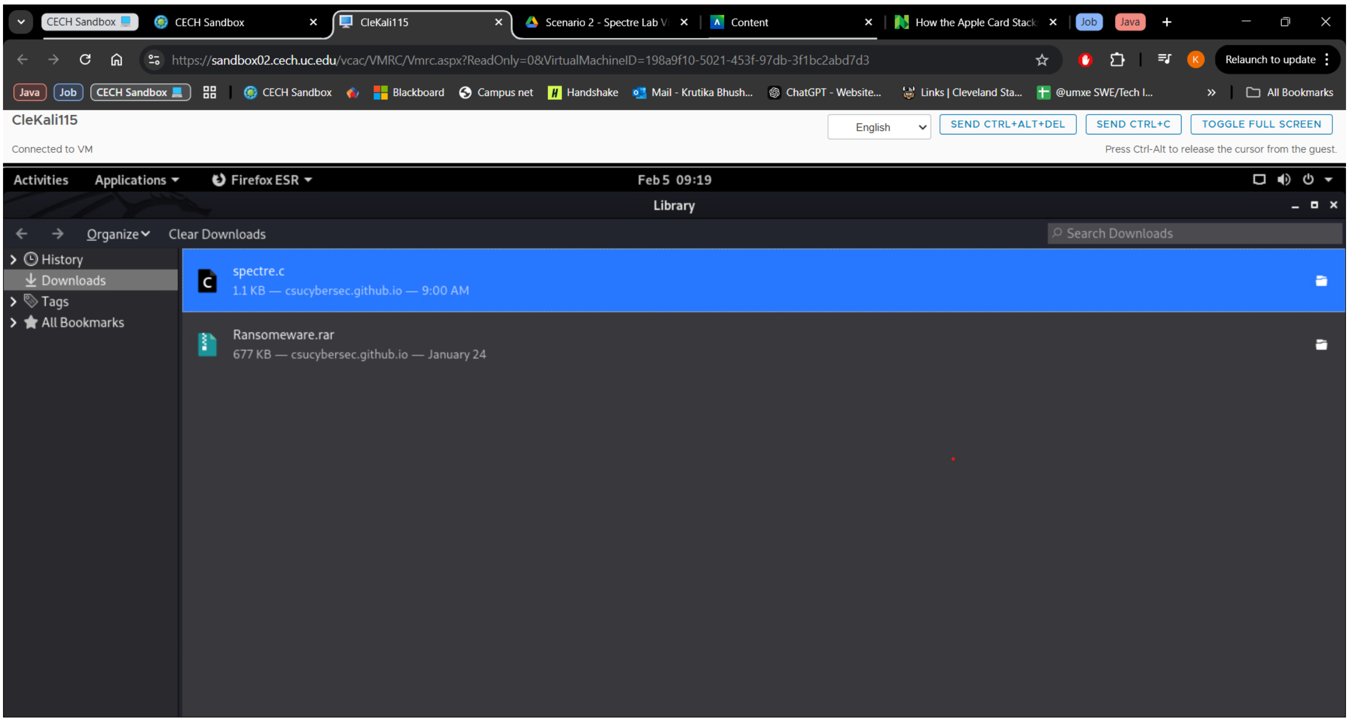
1. Provide the screenshot of the downloaded 'C' file.



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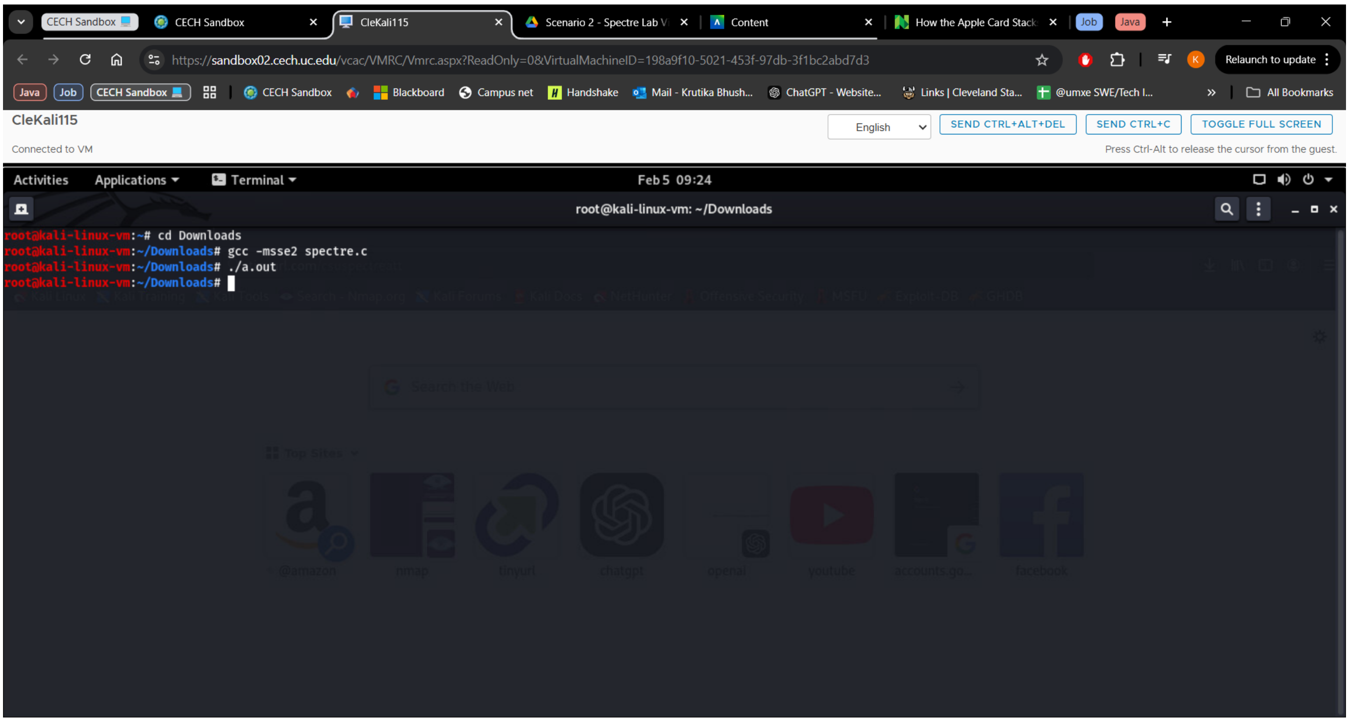
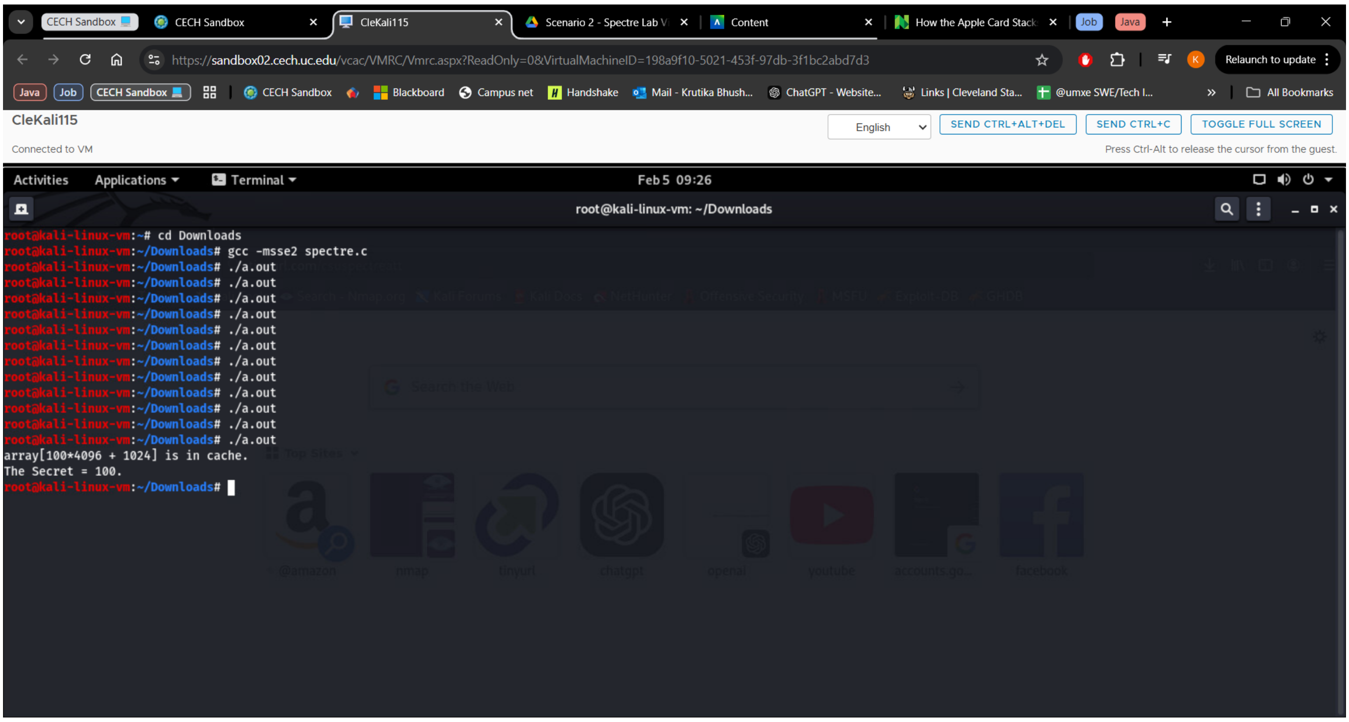
2. Provide the screenshot of the compilation and execution of the 'c' code.



3.Provide the screenshot of the desired output showing the secret value.



Exercise 2: Lab_Port Scanning

Part One: Exploring Kali Linux

Look through the applications installed on your Kali Linux System.

1. List and identify any tools that you recognize.

- **Nmap (Network Mapper)** – A network scanning tool used for discovering hosts and services on a network, identifying open ports, and performing security audits.
- **Medusa** – A brute-force password-cracking tool that can perform dictionary attacks on various protocols, such as SSH, FTP, and HTTP.
- **Wireshark** – A network protocol analyzer used for packet capturing and network traffic analysis. It helps in troubleshooting network issues and detecting security threats.
- **Sqlmap** – An automated SQL injection tool used for testing and exploiting SQL injection vulnerabilities in web applications.

Open the terminal application, type "nmap" and look at the man page for nmap. Read through this page and answer the following questions:

2. According to the description, what tasks do system and network administrators use nmap for?

Nmap (Network Mapper) is a powerful tool used by system and network administrators for various tasks, including:

1. Network Discovery – Finding active hosts and devices in a network.
2. Port Scanning – Identifying open ports on a target system to determine services running.
3. Service and Version Detection – Determining the exact software running on open ports.
4. OS Detection – Identifying the operating system running on a target host.
5. Security Auditing – Detecting vulnerabilities, misconfigurations, and potential security risks.
6. Firewall Evasion and IDS Testing – Testing firewall rules and Intrusion Detection System (IDS) capabilities.
7. Performance Monitoring – Checking server uptime and response times.

3. What option would you use to treat all hosts as online (skip host discovery)?

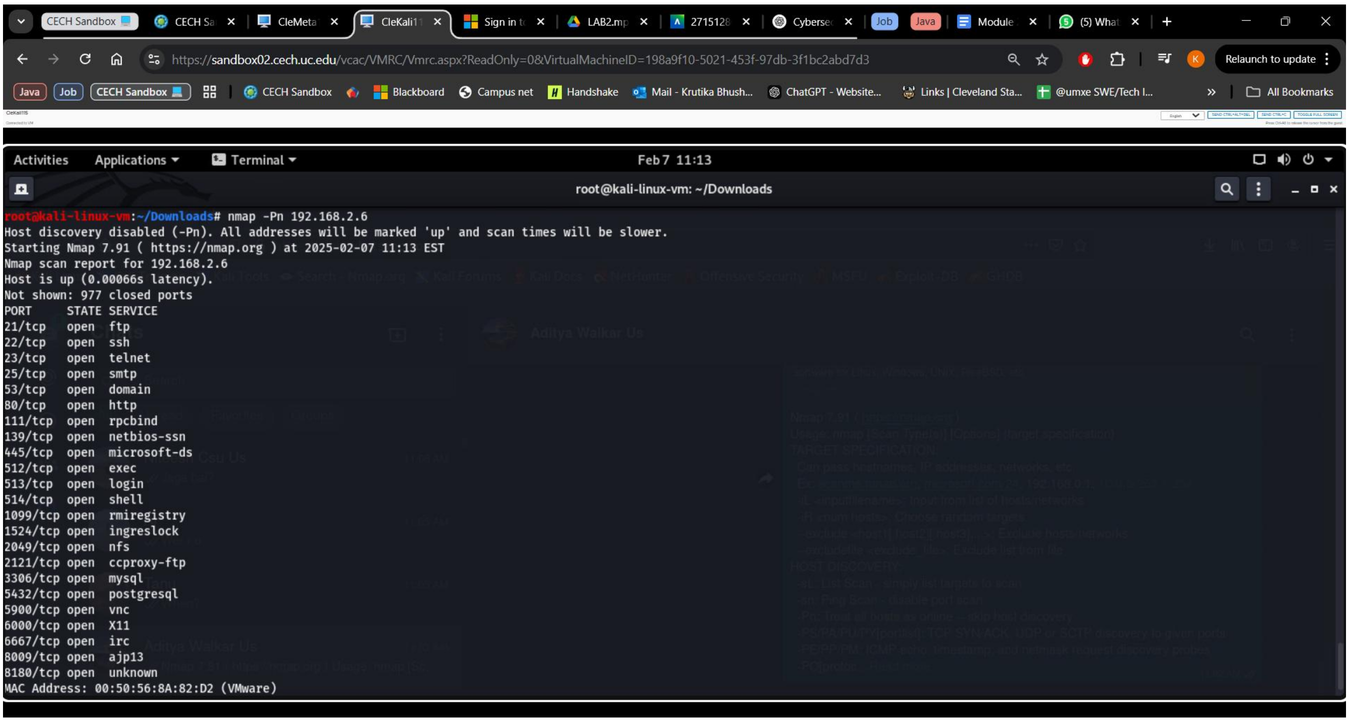
Option: -Pn

- By default, Nmap performs host discovery (pings the host before scanning).
- The -Pn option skips this step and treats all specified hosts as online.

Use case:

- Useful when scanning firewalled hosts that block ICMP ping requests.

nmap -Pn 192.168.2.6



4. What option would you use to specify specific ports to scan?

Option: -p <port ranges>

- This allows scanning specific ports rather than all 65535 ports.
- You can specify:
    - A single port: -p 22
    - A range of ports: -p 20-80
    - Multiple ports: -p 22,80,443

- UDP, TCP, and SCTP separately:
    - UDP: U:53
    - TCP: T:21-25,80
    - SCTP: S:9

Use case:

- When you only need to check if a particular service is running.

nmap -p 80,443 192.168.2.6

```
root@kali-linux-vm:~/Downloads# nmap -p 80,443 192.168.2.6
Starting Nmap 7.91 ( https://nmap.org ) at 2025-02-07 11:15 EST
Nmap scan report for 192.168.2.6
Host is up (0.00081s latency).

PORT    STATE  SERVICE
80/tcp  open   http
443/tcp closed https
MAC Address: 00:50:56:8A:82:D2 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.19 seconds
root@kali-linux-vm:~/Downloads#
```

5. What option would you use to determine service and version info on open ports?

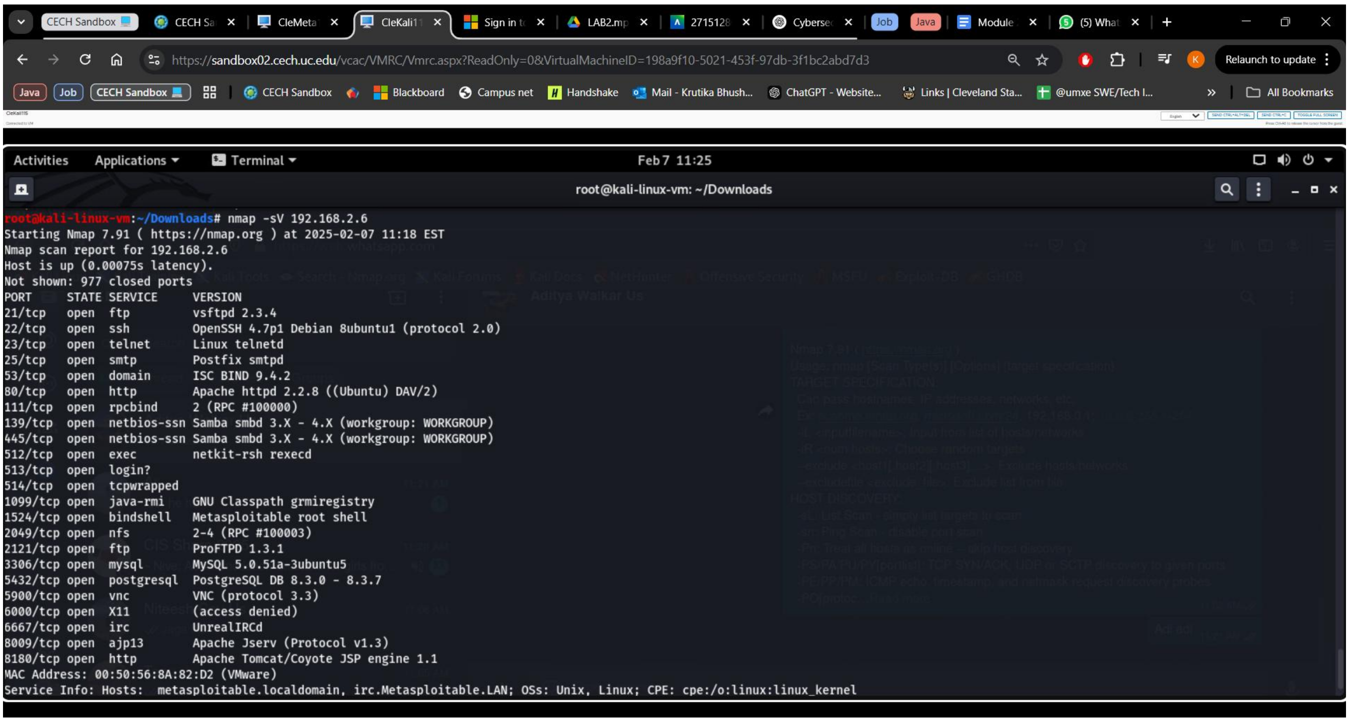
Option: -sV

- This probes open ports to identify running services and their versions.

Use case:

- Useful for penetration testing to check if outdated or vulnerable versions of software are running.

nmap -sV 192.168.2.6



6. What option would you use to enable OS detection?

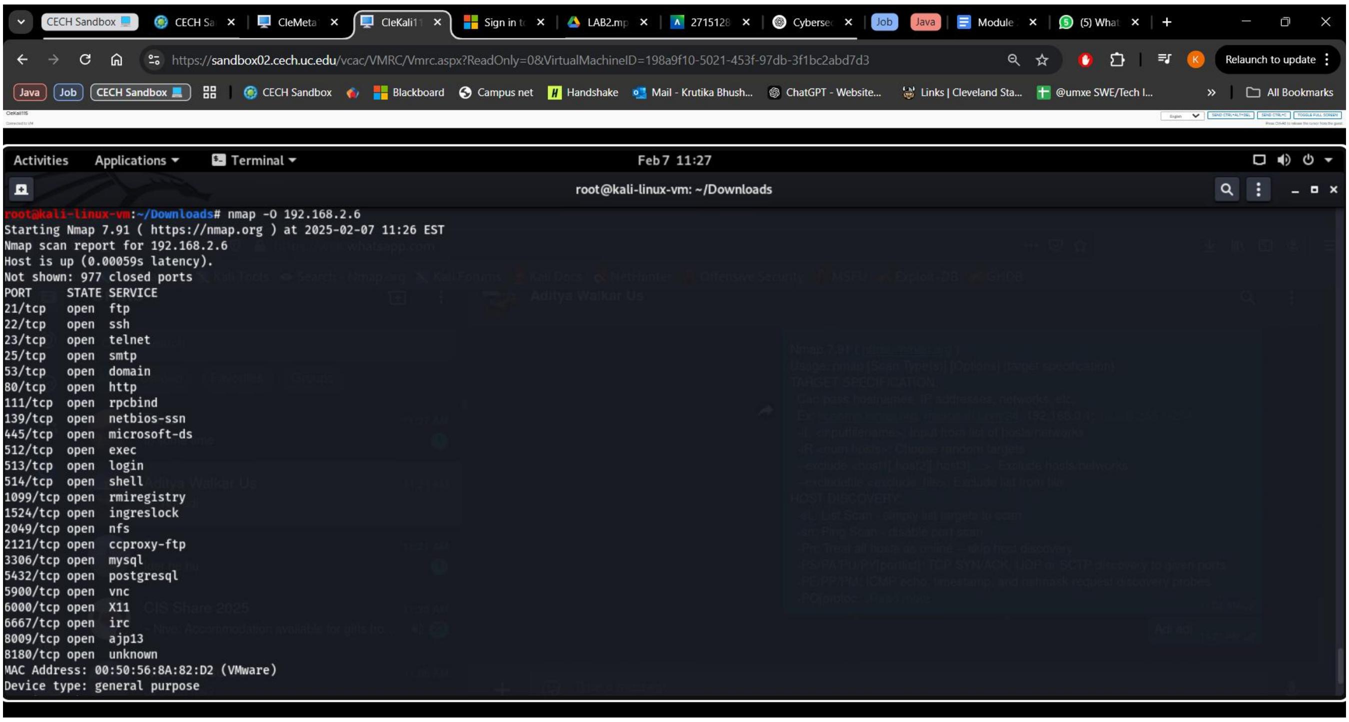
Option: -O

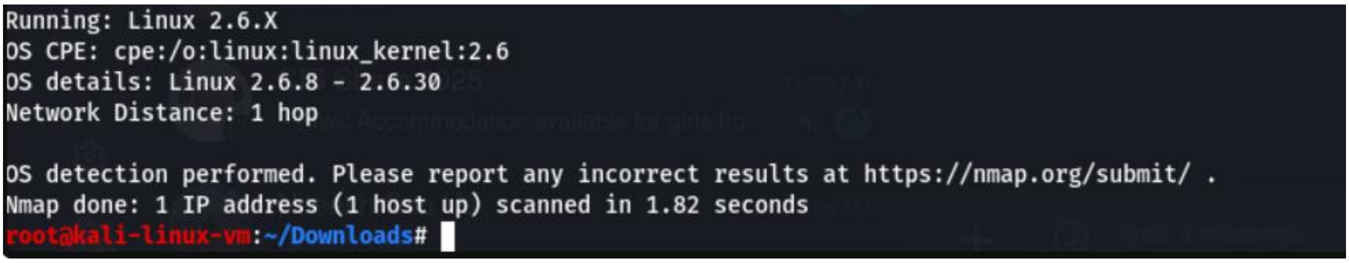
- This performs operating system detection using TCP/IP stack fingerprinting.

Use case:

- Helps in network security audits by identifying OS vulnerabilities.

nmap -O 192.168.2.6



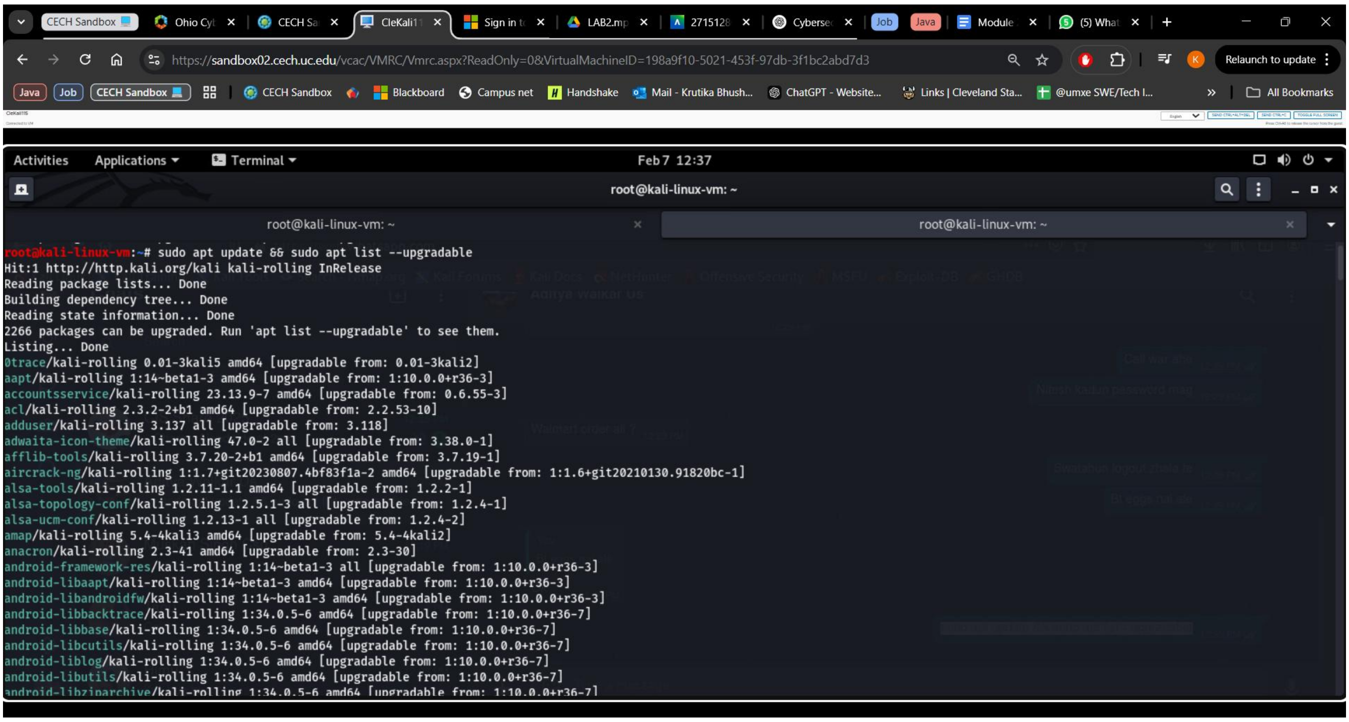


Check to see if you Kali system is up to date. If not, install all updates.

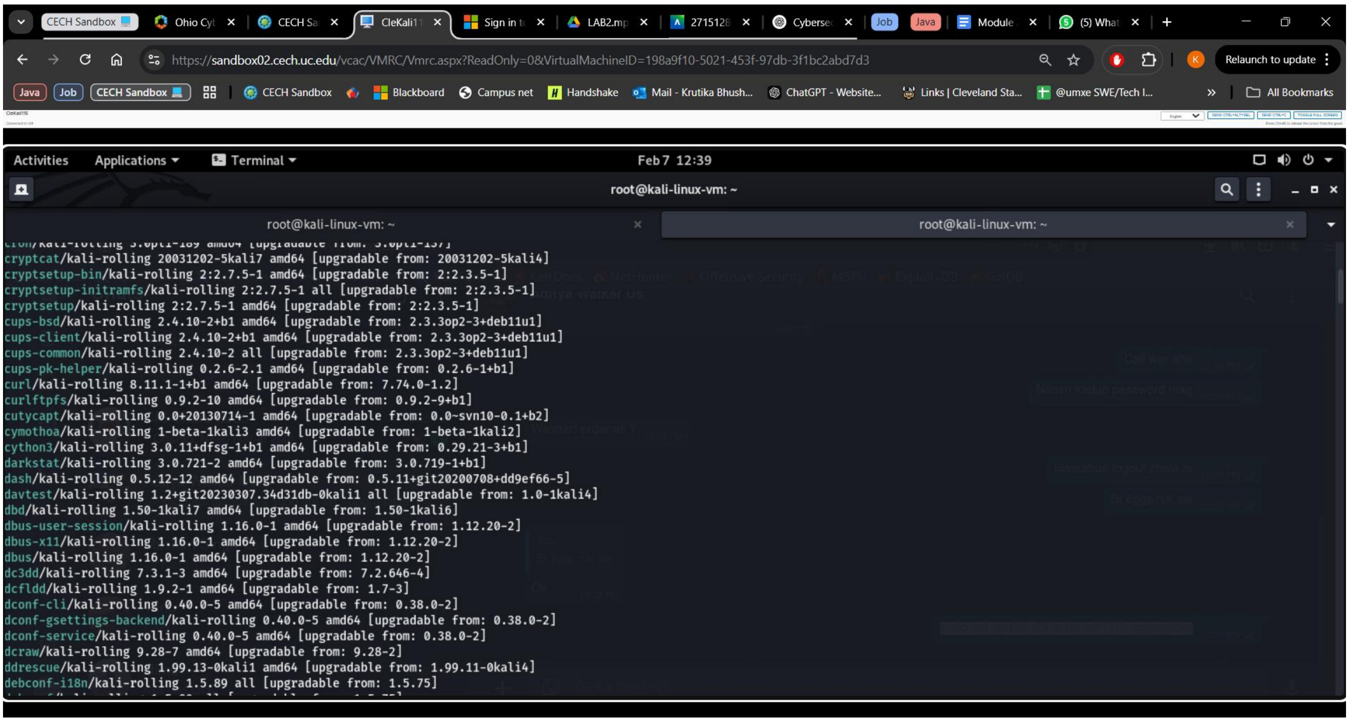
7. Paste a screen shot after updates are installed (or confirmed system is up to date.)

sudo apt update && sudo apt list –upgradable

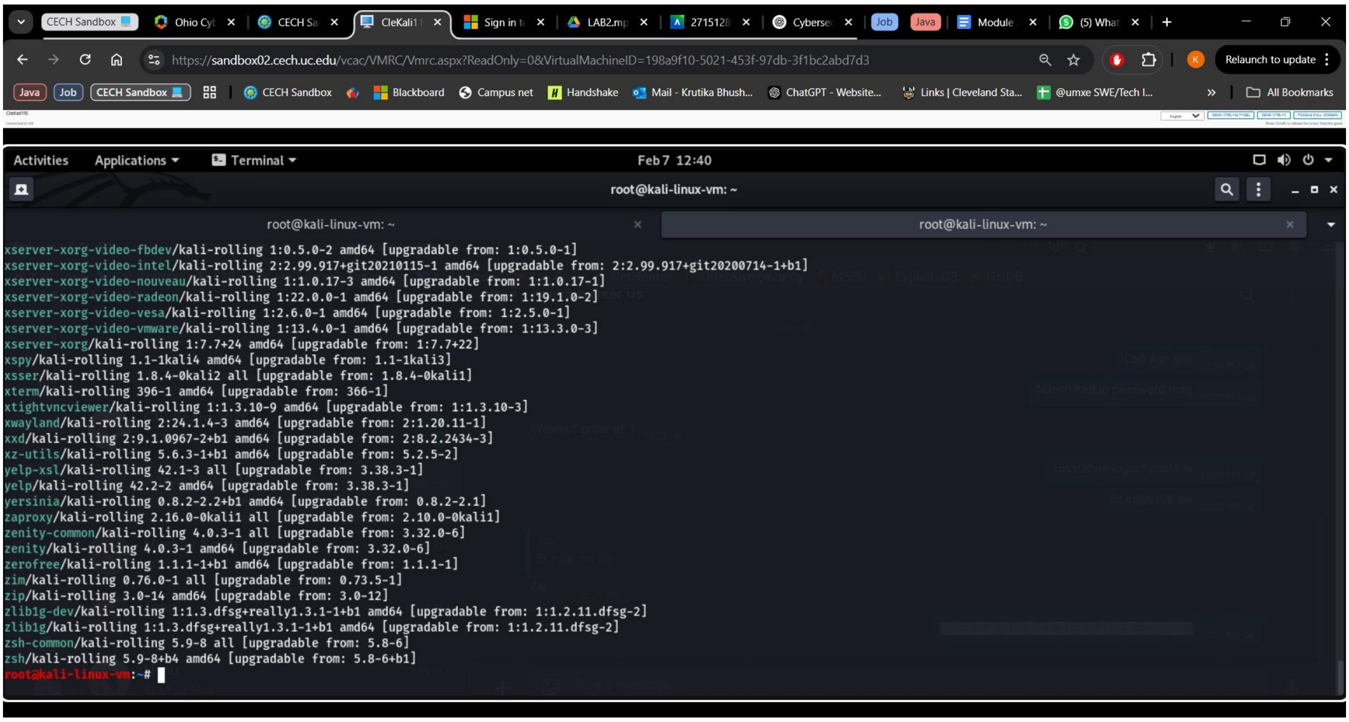
This command checks for updates

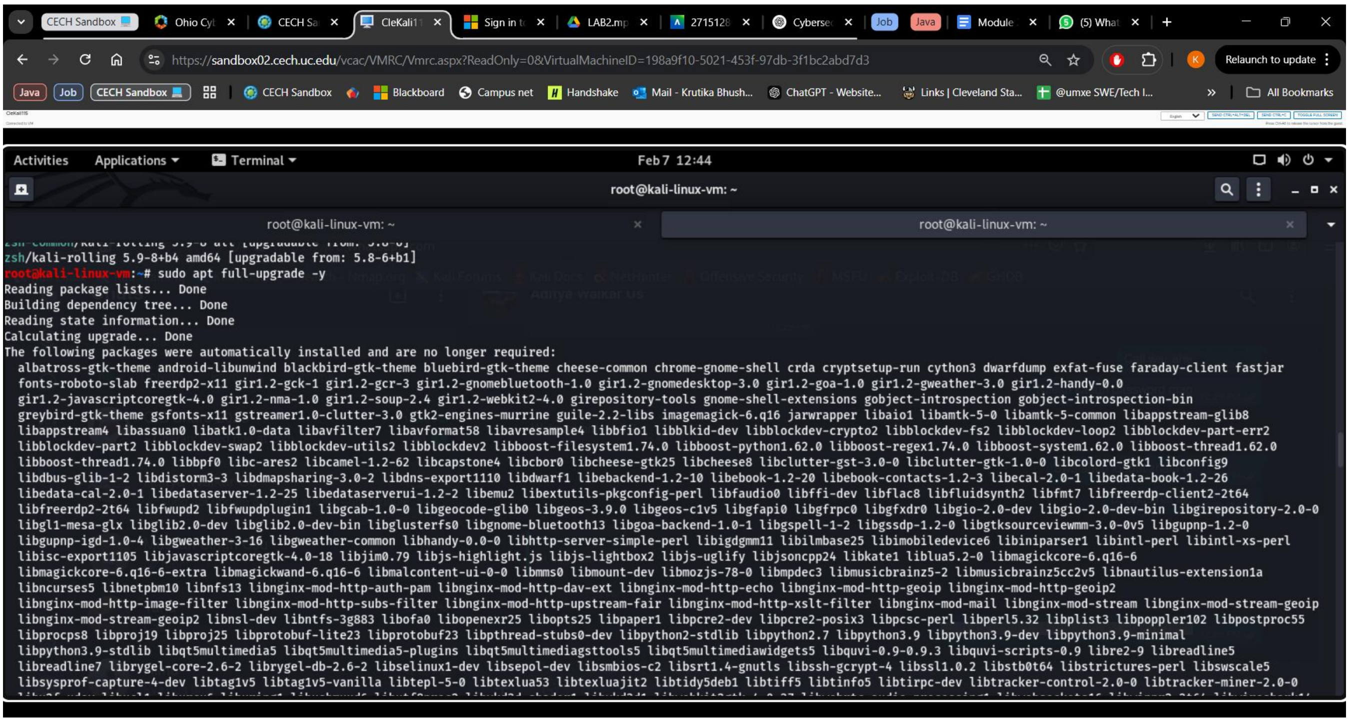


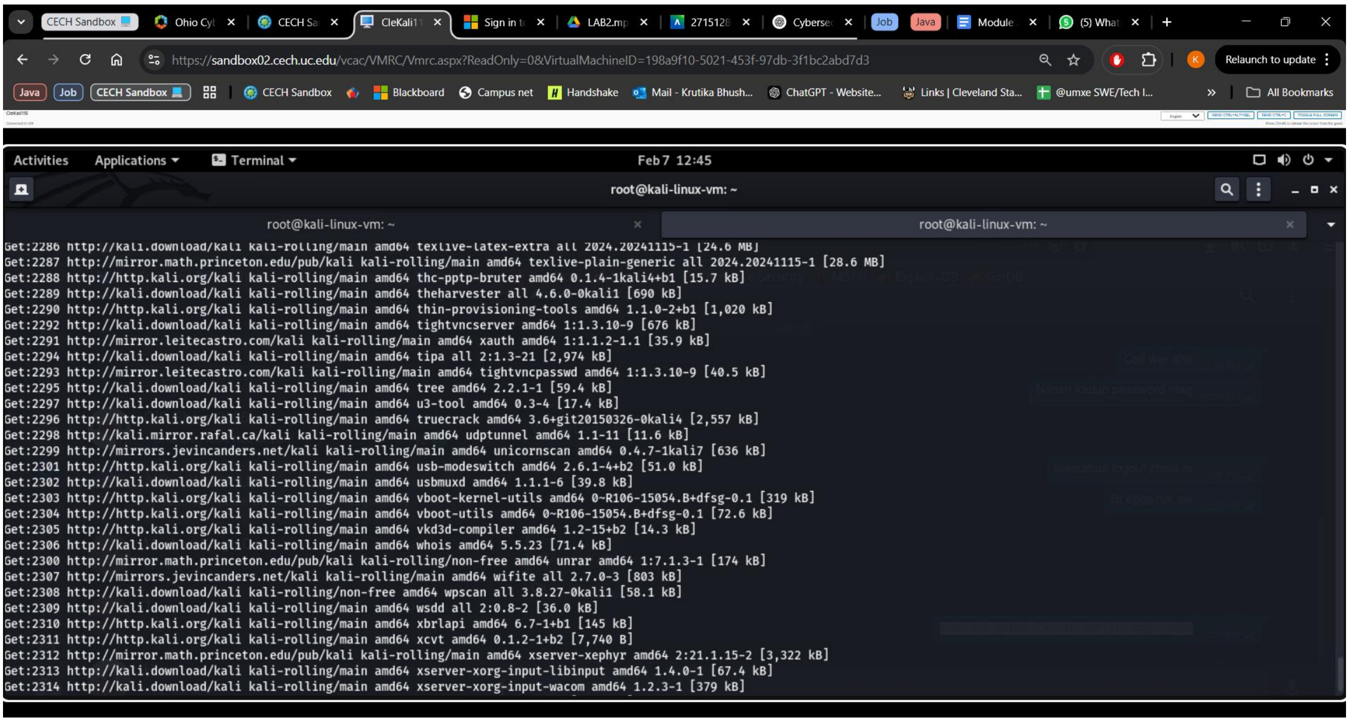
Activities
Applications
Terminal
Feb 7 12:37
root@kali-linux-vm: ~
root@kali-linux-vm:~# sudo apt update && sudo apt list --upgradable
Hit:1 http://http.kali.org/kali kali-rolling InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
2266 packages can be upgraded. Run 'apt list --upgradable' to see them.
Listing... Done
0trace/kali-rolling 0.01-3kali5 amd64 [upgradable from: 0.01-3kali2]
aapt/kali-rolling 1:14~beta1-3 amd64 [upgradable from: 1:10.0.0+r36-3]
accountsservice/kali-rolling 23.13.9-7 amd64 [upgradable from: 0.6.55-3]
acl/kali-rolling 2.3.2-2+b1 amd64 [upgradable from: 2.2.53-10]
adduser/kali-rolling 3.137 all [upgradable from: 3.118]
adwaita-icon-theme/kali-rolling 47.0-2 all [upgradable from: 3.38.0-1]
afflib-tools/kali-rolling 3.7.20-2+b1 amd64 [upgradable from: 3.7.19-1]
aircrack-ng/kali-rolling 1:1.7+git20230807.4bf83f1a-2 amd64 [upgradable from: 1:1.6+git20210130.91820bc-1]
alsa-tools/kali-rolling 1.2.11-1.1 amd64 [upgradable from: 1.2.2-1]
alsa-topology-conf/kali-rolling 1.2.5.1-3 all [upgradable from: 1.2.4-1]
alsa-ucm-conf/kali-rolling 1.2.13-1 all [upgradable from: 1.2.4-2]
amap/kali-rolling 5.4-4kali3 amd64 [upgradable from: 5.4-4kali2]
anacron/kali-rolling 2.3-41 amd64 [upgradable from: 2.3-30]
android-framework-res/kali-rolling 1:14~beta1-3 all [upgradable from: 1:10.0.0+r36-3]
android-libaapt/kali-rolling 1:14~beta1-3 amd64 [upgradable from: 1:10.0.0+r36-3]
android-libandroidfw/kali-rolling 1:14~beta1-3 amd64 [upgradable from: 1:10.0.0+r36-3]
android-libbacktrace/kali-rolling 1:34.0.5-6 amd64 [upgradable from: 1:10.0.0+r36-7]
android-libbase/kali-rolling 1:34.0.5-6 amd64 [upgradable from: 1:10.0.0+r36-7]
android-libcutils/kali-rolling 1:34.0.5-6 amd64 [upgradable from: 1:10.0.0+r36-7]
android-liblog/kali-rolling 1:34.0.5-6 amd64 [upgradable from: 1:10.0.0+r36-7]
android-libutils/kali-rolling 1:34.0.5-6 amd64 [upgradable from: 1:10.0.0+r36-7]
android-libziparchive/kali-rolling 1:34.0.5-6 amd64 [upgradable from: 1:10.0.0+r36-7]



Activities
Applications
Terminal
Feb 7 12:39
root@kali-linux-vm: ~
cryptcat/kali-rolling 20031202-5kali7 amd64 [upgradable from: 20031202-5kali4]
cryptsetup-bin/kali-rolling 2:2.7.5-1 amd64 [upgradable from: 2:2.3.5-1]
cryptsetup-initramfs/kali-rolling 2:2.7.5-1 all [upgradable from: 2:2.3.5-1]
cryptsetup/kali-rolling 2:2.7.5-1 amd64 [upgradable from: 2:2.3.5-1]
cups-bsd/kali-rolling 2.4.10-2+b1 amd64 [upgradable from: 2.3.3op2-3+deb11u1]
cups-client/kali-rolling 2.4.10-2+b1 amd64 [upgradable from: 2.3.3op2-3+deb11u1]
cups-common/kali-rolling 2.4.10-2 all [upgradable from: 2.3.3op2-3+deb11u1]
cups-pk-helper/kali-rolling 0.2.6-2.1 amd64 [upgradable from: 0.2.6-1+b1]
curl/kali-rolling 8.11.1-1+b1 amd64 [upgradable from: 7.74.0-1.2]
curlftpfs/kali-rolling 0.9.2-10 amd64 [upgradable from: 0.9.2-9+b1]
cutycapt/kali-rolling 0.0+20130714-1 amd64 [upgradable from: 0.0~svn10-0.1+b2]
cymothoa/kali-rolling 1-beta-1kali3 amd64 [upgradable from: 1-beta-1kali2]
cython3/kali-rolling 3.0.11+dfsg-1+b1 amd64 [upgradable from: 0.29.21-3+b1]
darkstat/kali-rolling 3.0.721-2 amd64 [upgradable from: 3.0.719-1+b1]
dash/kali-rolling 0.5.12-12 amd64 [upgradable from: 0.5.11+git20200708+dd9ef66-5]
davtest/kali-rolling 1.2+git20230307.34d31db-0kali1 all [upgradable from: 1.0-1kali4]
dbd/kali-rolling 1.50-1kali7 amd64 [upgradable from: 1.50-1kali6]
dbus-user-session/kali-rolling 1.16.0-1 amd64 [upgradable from: 1.12.20-2]
dbus-x11/kali-rolling 1.16.0-1 amd64 [upgradable from: 1.12.20-2]
dbus/kali-rolling 1.16.0-1 amd64 [upgradable from: 1.12.20-2]
dc3dd/kali-rolling 7.3.1-3 amd64 [upgradable from: 7.2.646-4]
dcfldd/kali-rolling 1.9.2-1 amd64 [upgradable from: 1.7-3]
dconf-cli/kali-rolling 0.40.0-5 amd64 [upgradable from: 0.38.0-2]
dconf-gsettings-backend/kali-rolling 0.40.0-5 amd64 [upgradable from: 0.38.0-2]
dconf-service/kali-rolling 0.40.0-5 amd64 [upgradable from: 0.38.0-2]
dcraw/kali-rolling 9.28-7 amd64 [upgradable from: 9.28-2]
ddrescue/kali-rolling 1.99.13-0kali1 amd64 [upgradable from: 1.99.11-0kali4]
debconf-i18n/kali-rolling 1.5.89 all [upgradable from: 1.5.75]


https://sandbox02.cech.uc.edu/vcac/VMRC/Vmrc.aspx?ReadOnly=0&VirtualMachineID=198a9f10-5021-453f-97db-3f1bc2abd7d3
Activities Applications Terminal
Feb 7 12:40
root@kali-linux-vm: ~
xserver-xorg-video-fbdev/kali-rolling 1:0.5.0-2 amd64 [upgradable from: 1:0.5.0-1]
xserver-xorg-video-intel/kali-rolling 2:2.99.917+git20210115-1 amd64 [upgradable from: 2:2.99.917+git20200714-1+b1]
xserver-xorg-video-nouveau/kali-rolling 1:1.0.17-3 amd64 [upgradable from: 1:1.0.17-1]
xserver-xorg-video-radeon/kali-rolling 1:22.0.0-1 amd64 [upgradable from: 1:19.1.0-2]
xserver-xorg-video-vesa/kali-rolling 1:2.6.0-1 amd64 [upgradable from: 1:2.5.0-1]
xserver-xorg-video-vmware/kali-rolling 1:13.4.0-1 amd64 [upgradable from: 1:13.3.0-3]
xserver-xorg/kali-rolling 1:7.7+24 amd64 [upgradable from: 1:7.7+22]
xspy/kali-rolling 1.1-1kali4 amd64 [upgradable from: 1.1-1kali3]
xsser/kali-rolling 1.8.4-0kali2 all [upgradable from: 1.8.4-0kali1]
xterm/kali-rolling 396-1 amd64 [upgradable from: 366-1]
xtightvncviewer/kali-rolling 1:1.3.10-9 amd64 [upgradable from: 1:1.3.10-3]
xwayland/kali-rolling 2:24.1.4-3 amd64 [upgradable from: 2:1.20.11-1]
xxd/kali-rolling 2:9.1.0967-2+b1 amd64 [upgradable from: 2:8.2.2434-3]
xz-utils/kali-rolling 5.6.3-1+b1 amd64 [upgradable from: 5.2.5-2]
yelp-xsl/kali-rolling 42.1-3 all [upgradable from: 3.38.3-1]
yelp/kali-rolling 42.2-2 amd64 [upgradable from: 3.38.3-1]
yersinia/kali-rolling 0.8.2-2.2+b1 amd64 [upgradable from: 0.8.2-2.1]
zaproxy/kali-rolling 2.16.0-0kali1 all [upgradable from: 2.10.0-0kali1]
zenity-common/kali-rolling 4.0.3-1 all [upgradable from: 3.32.0-6]
zenity/kali-rolling 4.0.3-1 amd64 [upgradable from: 3.32.0-6]
zerofree/kali-rolling 1.1.1-1+b1 amd64 [upgradable from: 1.1.1-1]
zim/kali-rolling 0.76.0-1 all [upgradable from: 0.73.5-1]
zip/kali-rolling 3.0-14 amd64 [upgradable from: 3.0-12]
zlib1g-dev/kali-rolling 1:1.3.dfsg+really1.3.1-1+b1 amd64 [upgradable from: 1:1.2.11.dfsg-2]
zlib1g/kali-rolling 1:1.3.dfsg+really1.3.1-1+b1 amd64 [upgradable from: 1:1.2.11.dfsg-2]
zsh-common/kali-rolling 5.9-8 all [upgradable from: 5.8-6]
zsh/kali-rolling 5.9-8+b4 amd64 [upgradable from: 5.8-6+b1]
root@kali-linux-vm:~#

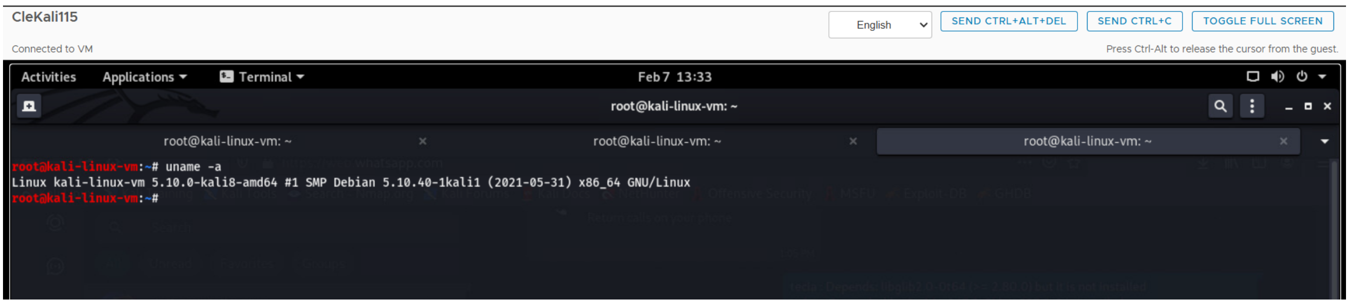

https://sandbox02.cech.uc.edu/vcac/VMRC/Vmrc.aspx?ReadOnly=0&VirtualMachineID=198a9f10-5021-453f-97db-3f1bc2abd7d3
Activities Applications Terminal
Feb 7 12:44
root@kali-linux-vm: ~
zsh/kali-rolling 5.9-8+b4 amd64 [upgradable from: 5.8-6+b1]
root@kali-linux-vm:~# sudo apt full-upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
albatross-gtk-theme android-libunwind blackbird-gtk-theme bluebird-gtk-theme cheese-common chrome-gnome-shell crda cryptsetup-run cython3 dwarfdump exfat-fuse faraday-client fastjar
fonts-roboto-slab freerdp2-x11 gir1.2-gck-1 gir1.2-gcr-3 gir1.2-gnomebluetooth-1.0 gir1.2-gnomedesktop-3.0 gir1.2-goa-1.0 gir1.2-gweather-3.0 gir1.2-handy-0.0
gir1.2-javascriptcoregtk-4.0 gir1.2-nma-1.0 gir1.2-soup-2.4 gir1.2-webkit2-4.0 girepository-tools gnome-shell-extensions gobject-introspection gobject-introspection-bin
greybird-gtk-theme gsfonts-x11 gstreamer1.0-clutter-3.0 gtk2-engines-murrine guile-2.2-libs imagemagick-6.q16 jarwrapper libaio1 libamtk-5-0 libamtk-5-common libappstream-glib8
libappstream4 libassuan0 libatk1.0-data libavfilter7 libavformat58 libavresample4 libbfio1 libblkid-dev libblockdev-crypto2 libblockdev-fs2 libblockdev-loop2 libblockdev-part-err2
libblockdev-part2 libblockdev-swap2 libblockdev-utils2 libblockdev2 libboost-filesystem1.74.0 libboost-python1.62.0 libboost-regex1.74.0 libboost-system1.62.0 libboost-thread1.62.0
libboost-thread1.74.0 libbpf0 libc-ares2 libcamel-1.2-62 libcapstone4 libcbor0 libcheese-gtk25 libcheese8 libclutter-gst-3.0-0 libclutter-gtk-1.0-0 libcolord-gtk1 libconfig9
libdbus-glib-1-2 libdistorm3-3 libdmapsharing-3.0-2 libdns-export1110 libdwarf1 libebackend-1.2-10 libebook-1.2-20 libebook-contacts-1.2-3 libecal-2.0-1 libedata-book-1.2-26
libedata-cal-2.0-1 libedataserver-1.2-25 libedataserverui-1.2-2 libemu2 libextutils-pkgconfig-perl libfaudio0 libffi-dev libflac8 libfluidsynth2 libfmt7 libfreerdp-client2-2t64
libfreerdp2-2t64 libfwupd2 libfwupdplugin1 libgcab-1.0-0 libgeocode-glib0 libgeos-3.9.0 libgeos-c1v5 libgfapi0 libgfrpc0 libgfxdr0 libgio-2.0-dev libgio-2.0-dev-bin libgirepository-2.0-0
libgl1-mesa-glx libglib2.0-dev libglib2.0-dev-bin libglusterfs0 libgnome-bluetooth13 libgoa-backend-1.0-1 libgspell-1-2 libgssdp-1.2-0 libgtksourceviewmm-3.0-0v5 libgupnp-1.2-0
libgupnp-igd-1.0-4 libgweather-3-16 libgweather-common libhandy-0.0-0 libhttp-server-simple-perl libigdgmm11 libilmbase25 libimobiledevice6 libiniparser1 libintl-perl libintl-xs-perl
libisc-export1105 libjavascriptcoregtk-4.0-18 libjim0.79 libjs-highlight.js libjs-lightbox2 libjs-uglify libjsoncpp24 libkate1 liblua5.2-0 libmagickcore-6.q16-6
libmagickcore-6.q16-6-extra libmagickwand-6.q16-6 libmalcontent-ui-0-0 libmms0 libmount-dev libmozjs-78-0 libmpdec3 libmusicbrainz5-2 libmusicbrainz5cc2v5 libnautilus-extension1a
libncurses5 libnetpbm10 libnfs13 libnginx-mod-http-auth-pam libnginx-mod-http-dav-ext libnginx-mod-http-echo libnginx-mod-http-geoip libnginx-mod-http-geoip2
libnginx-mod-http-image-filter libnginx-mod-http-subs-filter libnginx-mod-http-upstream-fair libnginx-mod-http-xslt-filter libnginx-mod-mail libnginx-mod-stream libnginx-mod-stream-geoip
libnginx-mod-stream-geoip2 libnsl-dev libntfs-3g883 libofa0 libopenexr25 libopts25 libpaper1 libpcre2-dev libpcre2-posix3 libpcsc-perl libperl5.32 libplist3 libpoppler102 libpostproc55
libprocps8 libproj19 libproj25 libprotobuf-lite23 libprotobuf23 libpthread-stubs0-dev libpython2-stdlib libpython2.7 libpython3.9 libpython3.9-dev libpython3.9-minimal
libpython3.9-stdlib libqt5multimedia5 libqt5multimedia5-plugins libqt5multimediagsttools5 libqt5multimediawidgets5 libquvi-0.9-0.9.3 libquvi-scripts-0.9 libre2-9 libreadline5
libreadline7 librygel-core-2.6-2 librygel-db-2.6-2 libselinux1-dev libsepol-dev libsmbios-c2 libsrt1.4-gnutls libssh-gcrypt-4 libssl1.0.2 libstb0t64 libstrictures-perl libswscale5
libsysprof-capture-4-dev libtag1v5 libtag1v5-vanilla libtepl-5-0 libtexlua53 libtexluajit2 libtidy5deb1 libtiff5 libtinfo5 libtirpc-dev libtracker-control-2.0-0 libtracker-miner-2.0-0



8. What distribution of Linux is Kali based on? (Hint: use the uname command)

Kali Linux is based on Debian.

uname -a command output includes "Debian", which confirms that Kali is a Debian-based distribution. It follows a rolling release model and is specifically tailored for penetration testing, cybersecurity, and digital forensics.



Part Two: Starting Metasploit Framework

Use the Metasploit Unleashed site for help completing this portion of the lab.

9. In your own words, explain what the Meatasploit Framework is.

The Metasploit Framework is an open-source cybersecurity tool used for penetration testing and security assessments. It allows ethical hackers and security professionals to simulate real-world cyberattacks to identify and fix vulnerabilities in networks, applications, and devices. With a vast collection of exploits, payloads, and tools, Metasploit helps automate attacks and test defenses, ensuring that security weaknesses are addressed before malicious hackers can exploit them. It is widely used in the cybersecurity field to improve overall system security and resilience.

Start the postgresql, make sure the 'msfdb' is initialized, and open the msfconsole.

In Terminal type: services psotgresql start

In Terminal type: Msdb init

In Terminal type: msfconconsole

10. Paste as screen show of each of the commands used to complete these steps

Start the postgresql and check status

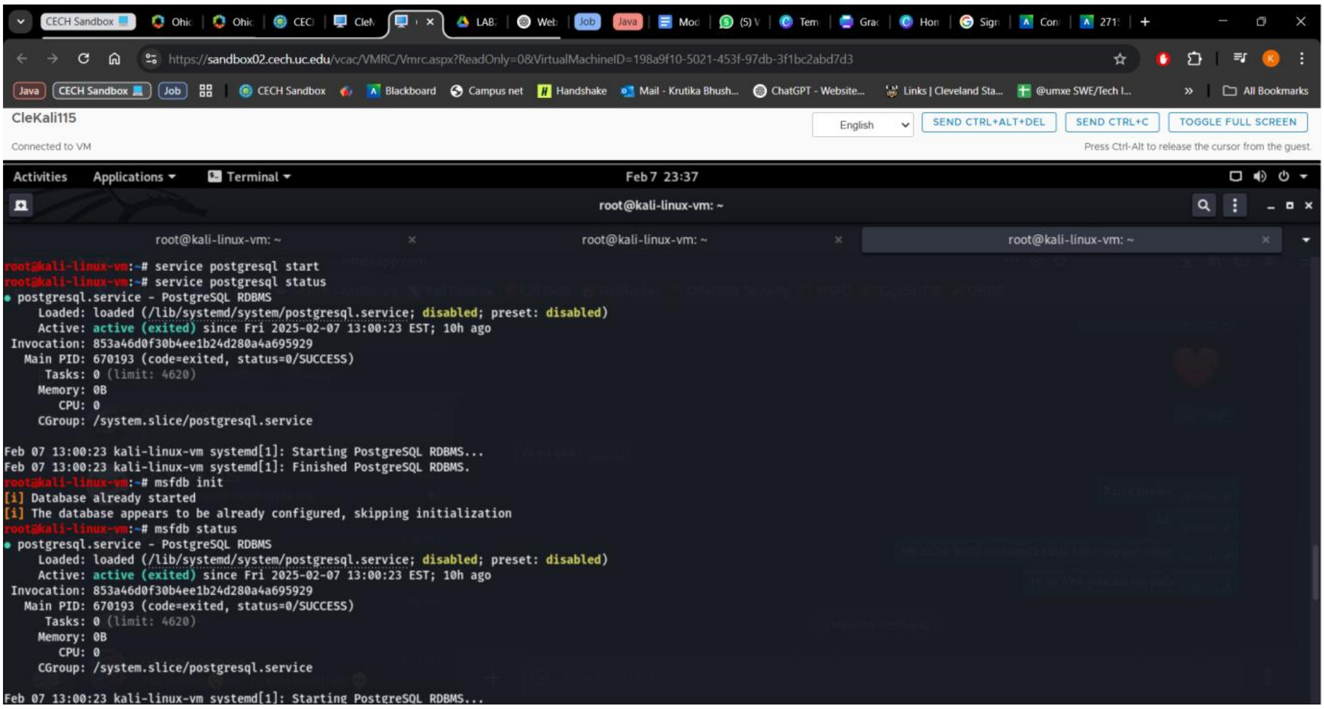
service postgresql start

service postgresql status

Initialize msfdb and check status

msfdb init

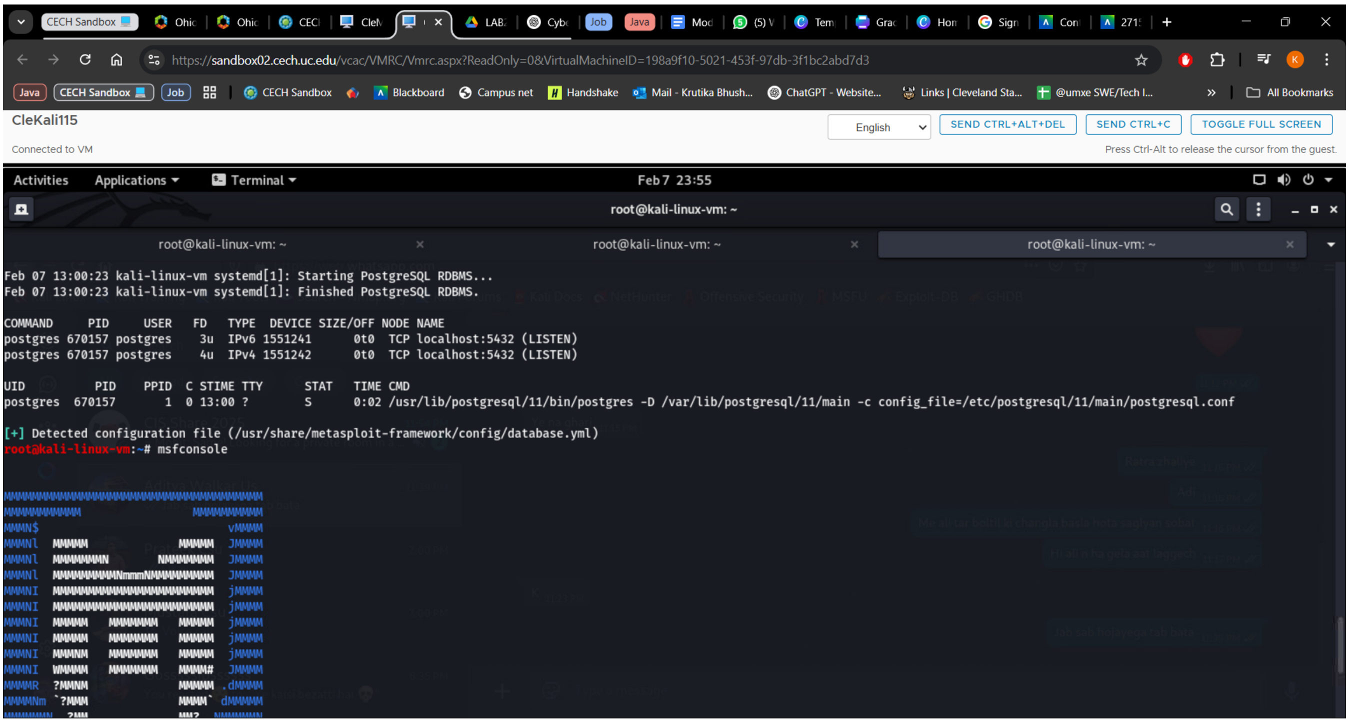
msfdb status

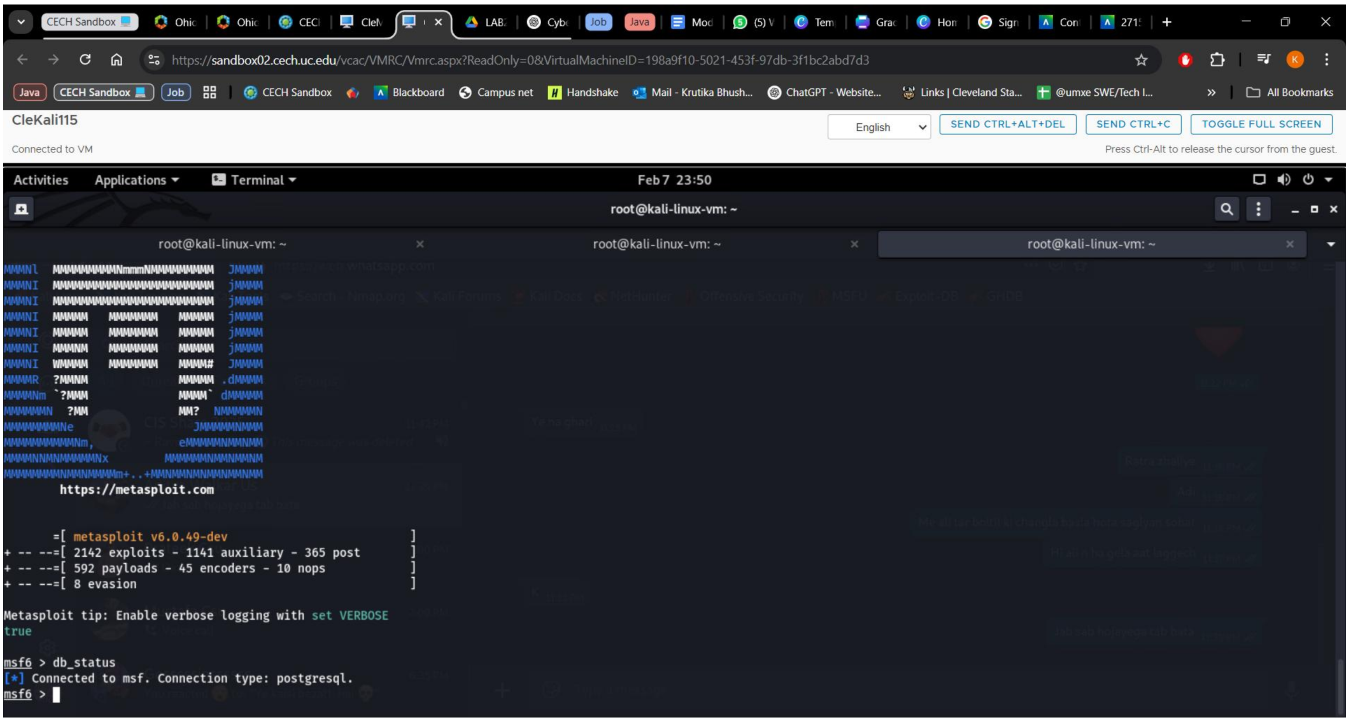


Starting the msfconsole and checking connection

msfconsole

db_status





11. What is the IP and subnet mask of your Kali Linux System?

The IP address of my Kali Linux system is **192.168.2.7**, and the subnet mask is **255.255.255.0**.

12. What is the Network Address of your network?

The network address of my network can be determined using the IP address and subnet mask. Since my IP is **192.168.2.7** and the subnet mask is **255.255.255.0**, the network address is **192.168.2.0**.

```
msf6 > ifconfig
[*] exec: ifconfig

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.2.7  netmask 255.255.255.0  broadcast 192.168.2.255
        inet6 fe80::250:56ff:fe8a:6cae  prefixlen 64  scopeid 0x20<link>
        ether 00:50:56:8a:6c:ae  txqueuelen 1000  (Ethernet)
        RX packets 832065  bytes 4138083411 (3.8 GiB)
        RX errors 0  dropped 28  overruns 0  frame 0
        TX packets 721243  bytes 59896842 (57.1 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 13490140  bytes 3183958312 (2.9 GiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 13490140  bytes 3183958312 (2.9 GiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

13. What hosts are listed in your database now? (Show a screen shot)

```
msf6 > hosts

Hosts
=====

address      mac  name             os_name    os_flavor   os_sp  purpose  info  comments
-------      ---  ----             -------    ---------   -----  -------  ----  --------
192.168.2.3       WIN-678LK3JBN3Q  Windows 7  Enterprise  SP1    client
192.168.2.6
```

14. What services are listed in your database now? (Show a screen shot)

```
msf6 > services
Services
========

host         port  proto  name  state  info
----         ----  -----  ----  -----  ----
192.168.2.3  445   tcp
```

Run an nmap ping sweep in your network. (nmap -sn 192.168.2.0/24)

15. Paste a screen show of the command and results.

```
msf6 > nmap -sn 192.168.2.6
[*] exec: nmap -sn 192.168.2.6

Starting Nmap 7.91 ( https://nmap.org ) at 2025-02-08 00:20 EST
Nmap scan report for 192.168.2.6
Host is up (0.0018s latency).
MAC Address: 00:50:56:8A:82:D2 (VMware)
Nmap done: 1 IP address (1 host up) scanned in 0.15 seconds
msf6 >
```

16. What is the IP address of you metasploit2 target? (Take note of this for future assignments)

The IP address for my metasploit2 target is 192.168.2.6

```
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:50:56:8a:82:d2
          inet addr:192.168.2.6  Bcast:192.168.2.255  Mask:255.255.255.0
          inet6 addr: fe80::250:56ff:fe8a:82d2/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:123778 errors:10 dropped:11 overruns:0 frame:0
          TX packets:107553 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:13970593 (13.3 MB)  TX bytes:25250358 (24.0 MB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:99427 errors:0 dropped:0 overruns:0 frame:0
          TX packets:99427 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:49623921 (47.3 MB)  TX bytes:49623921 (47.3 MB)
```

Run a nmap scan against your target. This time use db_nmap to store the results in your database. Specify options to meet the following criteria:

• Scan ports 22,53,80,443 and 55432

• Run OS detection

17. Show a screen show if the command and the results.

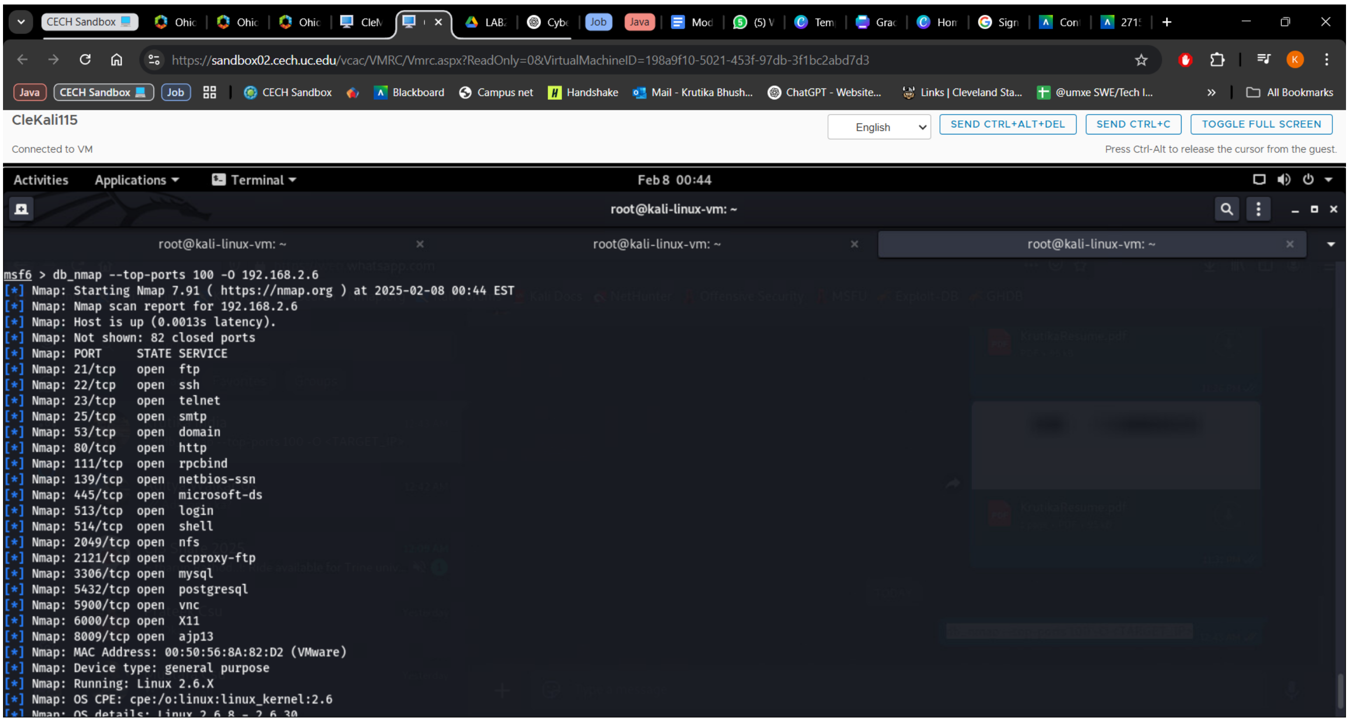
db_nmap → Saves results in the Metasploit database.
-p 22,53,80,443,55432 → Scans the specified ports.
-O → Enables OS detection to identify the target system's operating system.

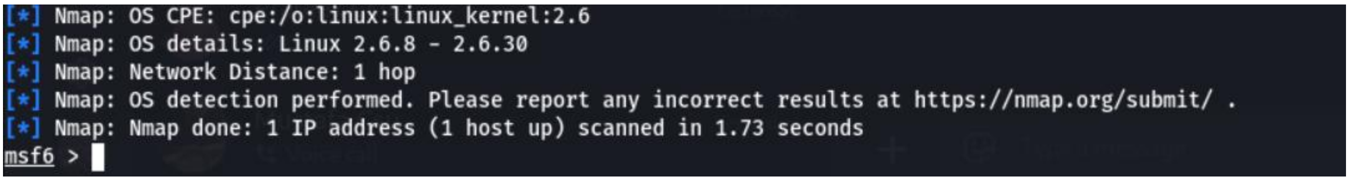
```
msf6 > db_nmap -p 22,53,80,443,55432 -O 192.168.2.6
[*] Nmap: Starting Nmap 7.91 ( https://nmap.org ) at 2025-02-08 00:37 EST
[*] Nmap: Nmap scan report for 192.168.2.6
[*] Nmap: Host is up (0.00093s latency).
[*] Nmap: PORT      STATE  SERVICE
[*] Nmap: 22/tcp    open   ssh
[*] Nmap: 53/tcp    open   domain
[*] Nmap: 80/tcp    open   http
[*] Nmap: 443/tcp   closed https
[*] Nmap: 55432/tcp closed unknown
[*] Nmap: MAC Address: 00:50:56:8A:82:D2 (VMware)
[*] Nmap: Device type: general purpose
[*] Nmap: Running: Linux 2.6.X
[*] Nmap: OS CPE: cpe:/o:linux:linux_kernel:2.6
[*] Nmap: OS details: Linux 2.6.8 - 2.6.30
[*] Nmap: Network Distance: 1 hop
[*] Nmap: OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
[*] Nmap: Nmap done: 1 IP address (1 host up) scanned in 1.74 seconds
```

18. What services are now listed in your database? (Show a screen shot) (Type: hosts)

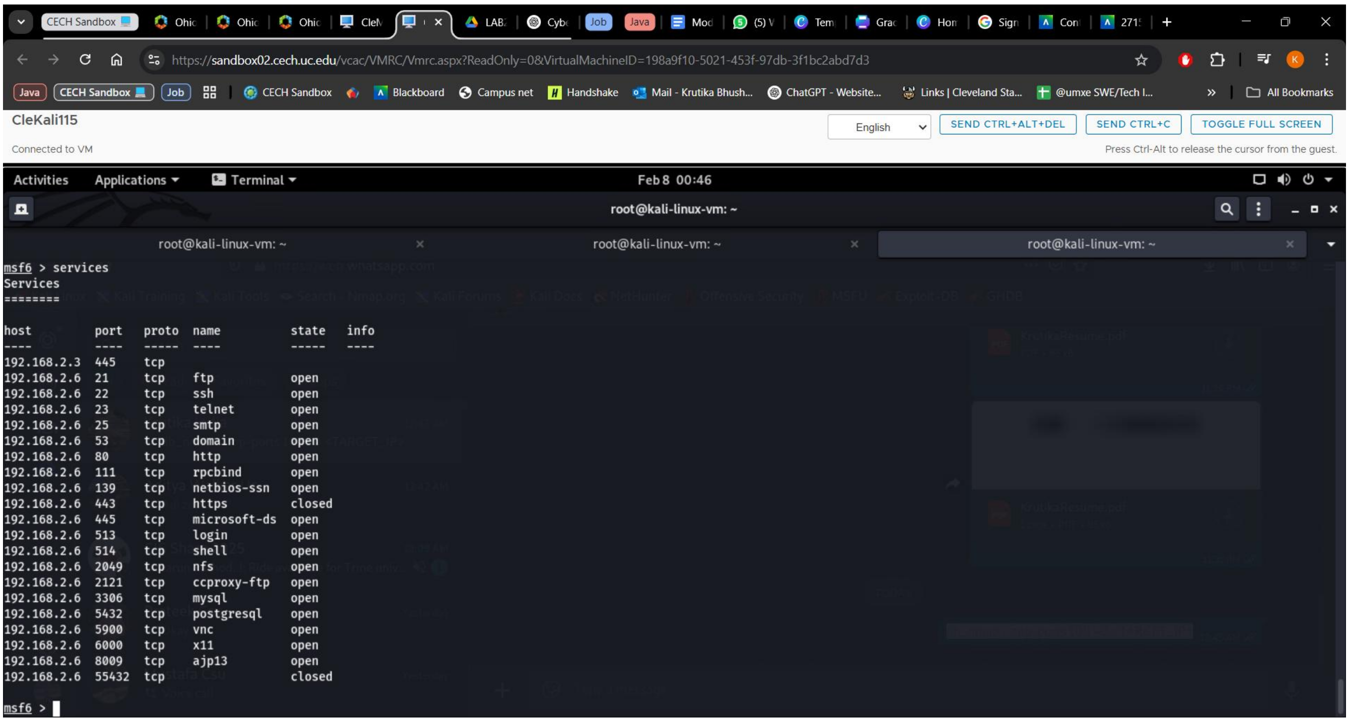
```
msf6 > hosts

Hosts
=====

address      mac                name             os_name    os_flavor   os_sp  purpose  info  comments
-------      ---                ----             -------    ---------   -----  -------  ----  --------
192.168.2.3                     WIN-678LK3JBN3Q  Windows 7  Enterprise  SP1    client
192.168.2.6  00:50:56:8a:82:d2                   Linux                  2.6.X  server

msf6 > services
Services
========

host         port   proto  name    state   info
----         ----   -----  ----    -----   ----
192.168.2.3  445    tcp
192.168.2.6  22     tcp    ssh     open
192.168.2.6  53     tcp    domain  open
192.168.2.6  80     tcp    http    open
192.168.2.6  443    tcp    https   closed
192.168.2.6  55432  tcp            closed

msf6 >
```

19. Run another nmap scan against your target, this time choose the top 100 ports. (Show a screen shot)





20. What services are running on your target? (Show a screen shot)



Run a scan with one of Metasploits built in port scanning tools against your target. You choose the tool and the options.

---

21. Paste a screen shot of the options you choose and the results of running the scanner

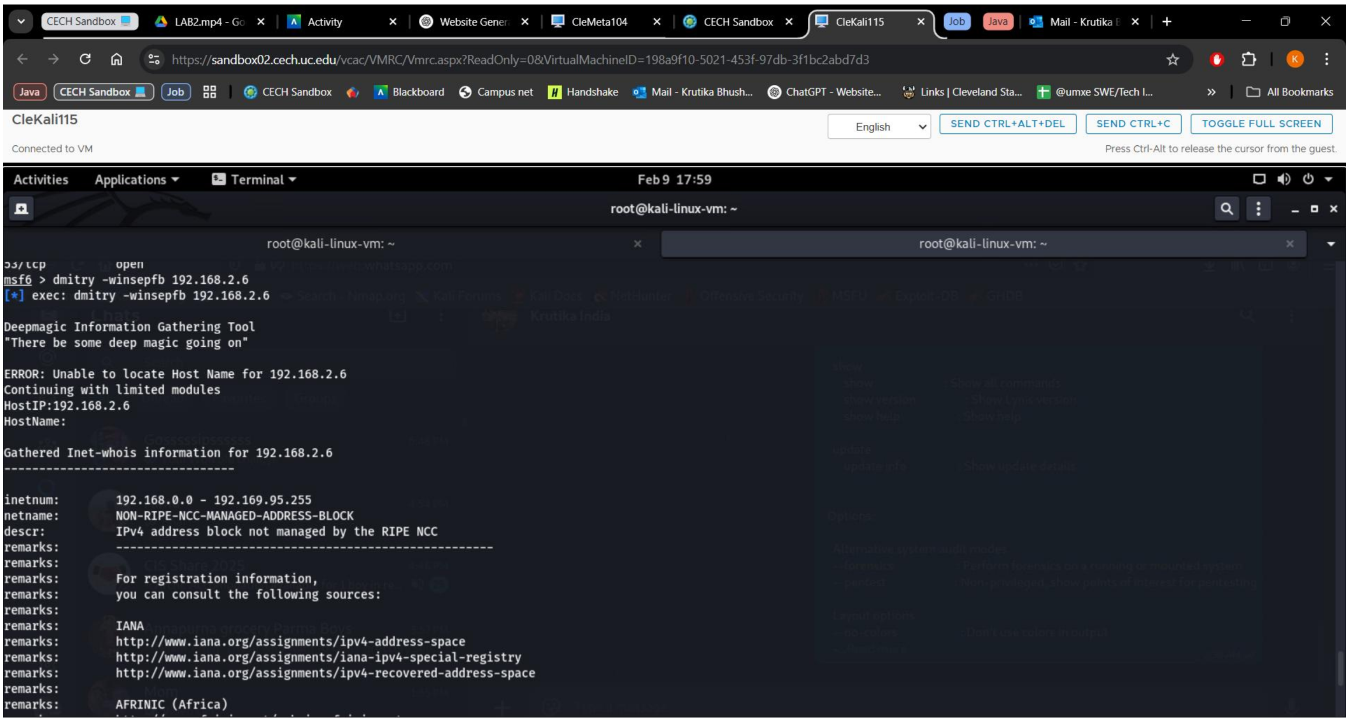
---

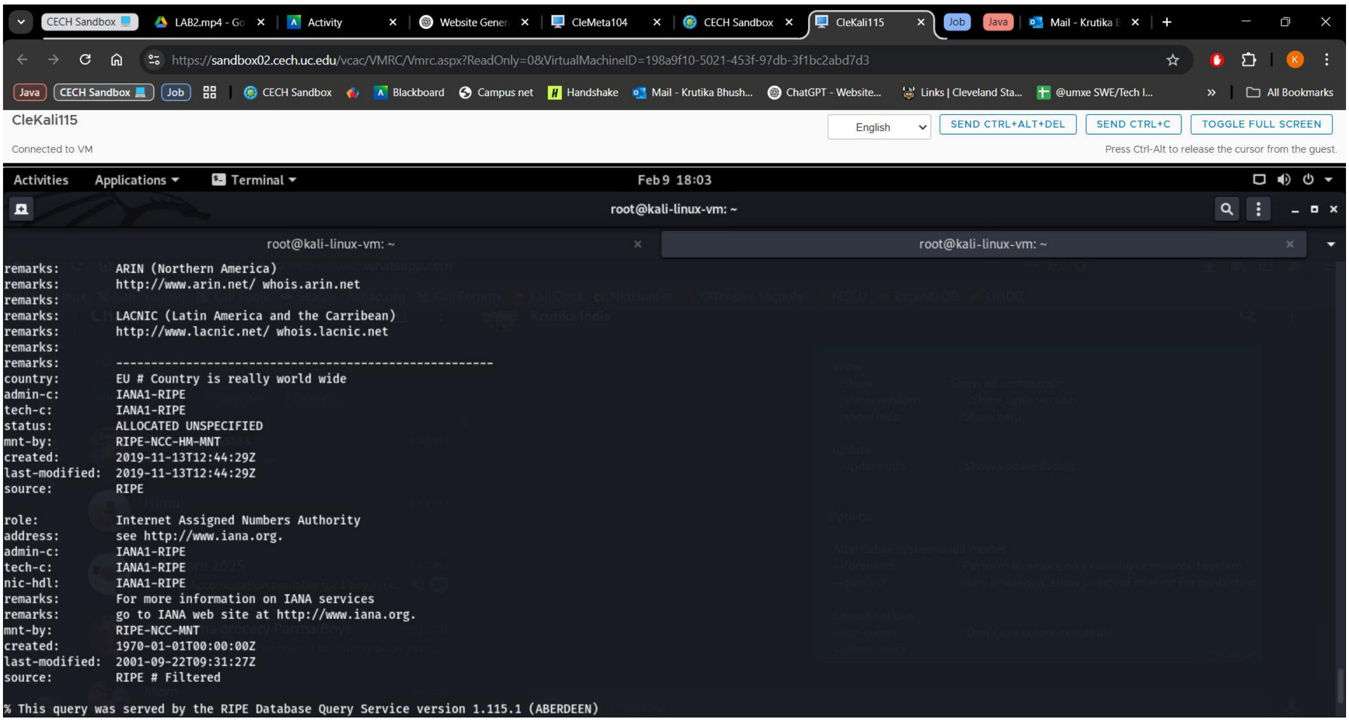
Dmitry -winsepfb 192.168.2.6

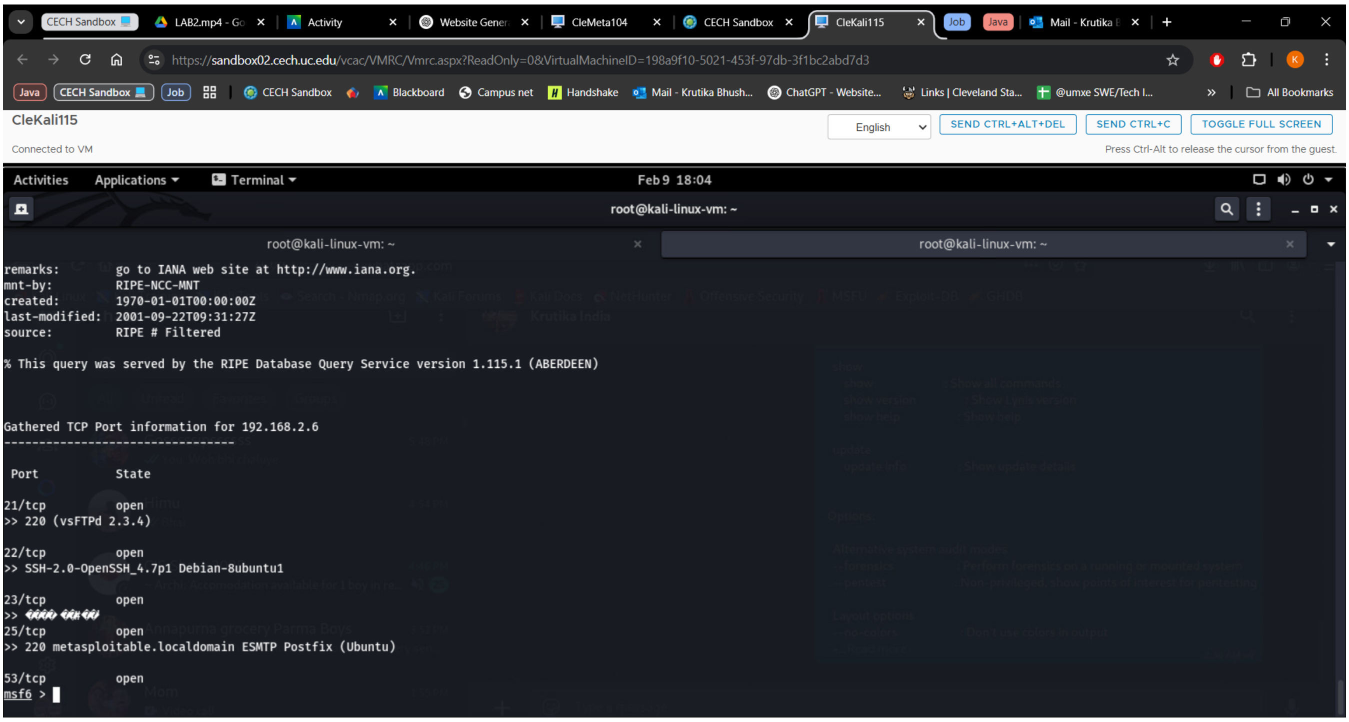
This command runs Dmitry (Deepmagic Information Gathering Tool) with multiple options enabled against the target IP 192.168.2.6.

Option	Description
-w	Perform a **whois lookup** on the target (both domain and IP-based).
-i	Retrieve **IP address** information for the target.
-n	Perform an **NS lookup** (Name Server lookup) to gather DNS details.
-s	Conduct a **subdomain search** to find related domains.
-e	Perform an **email address search** on the target domain.
-p	Scan **open ports** on the target system.
-f	Enable **full detailed output**, providing an extensive report.

Option	Description
-b	Conduct a **banner grabbing scan** to determine service versions.







22. Did you find any additional services that are running? List the service and ports below.

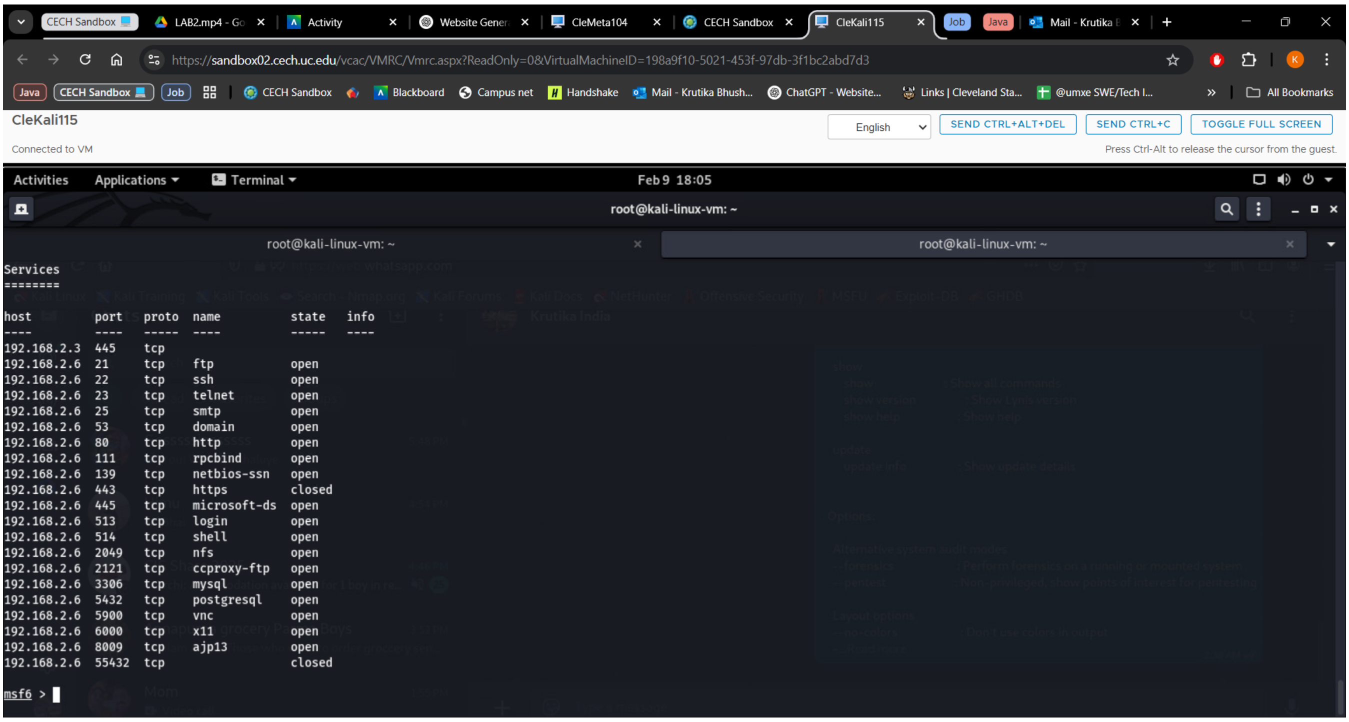
Based on the scan results, these additional services were found running on **192.168.2.6**:

Port	Protocol	Service	State
21	TCP	FTP	Open
22	TCP	SSH	Open
23	TCP	Telnet	Open
25	TCP	SMTP	Open
53	TCP	DNS	Open
80	TCP	HTTP	Open
111	TCP	RPCBind	Open
139	TCP	NetBIOS-SSN	Open
445	TCP	Microsoft-DS	Open

Port	Protocol	Service	State
513	TCP	Login	Open
514	TCP	Shell	Open
2049	TCP	NFS	Open
2121	TCP	CCProxy-FTP	Open
3306	TCP	MySQL	Open
5432	TCP	PostgreSQL	Open
5900	TCP	VNC	Open
6000	TCP	X11	Open
8009	TCP	AJP13	Open

Closed Ports:

- 443 (HTTPS)
- 55432 (Unknown Service)



Exit the msfconsole.

Part Four: Bonus

Choose one of the other port scanning tools discussed in the course materials (p0f, Xprobe2, Masscan, Netcat). Run any kind of scan against your target system that you like. Then record a short video (under 3 minutes) and explain what options you choose and describe the results.

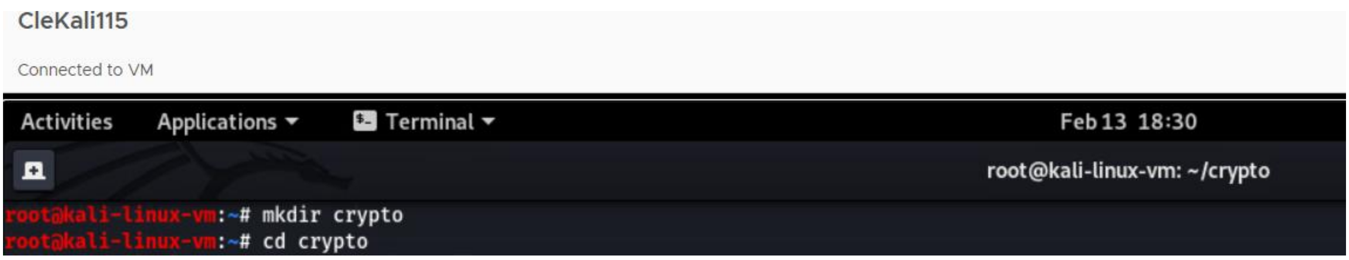
Attached Separately in the submission

Exercise 3: Lab_Cryptography (Cryptography, Encoding, and Numbering Systems)

Part Zero: Hashing

Open a terminal window on the Linux SIFT workstation. Create a folder and navigate to the newly created folder:

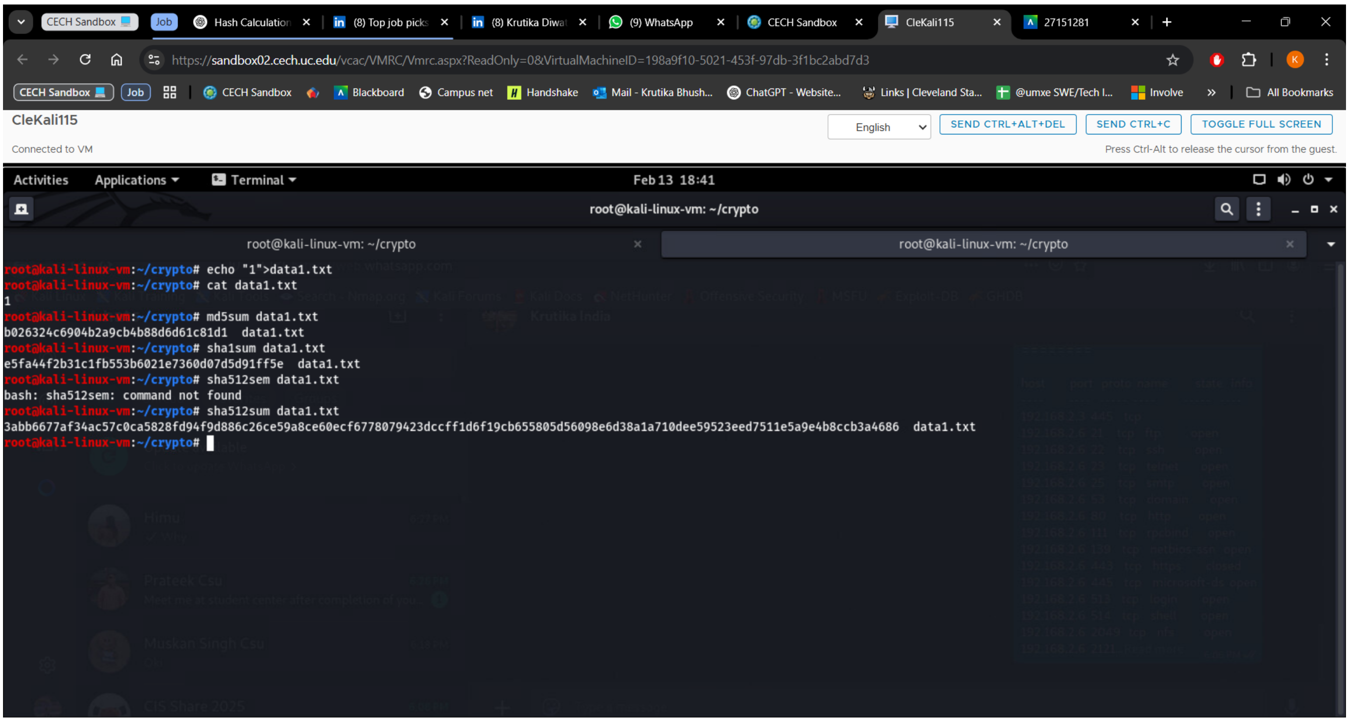
- **mkdir crypto**
- **cd crypto**



Create a file, calculate hashes using different algorithms. Compare the output hash values:

- **echo "1" > data1.txt**
- **cat data.txt**
- **md5sum data1.txt**
- **sha1sum data1.txt**

- **sha512sum data1.txt**



Create another file with the same content. Note that the filename does not change the hash value, but the file content does change:

- **echo "1" > data2.txt**
- **cat data2.txt**
- **md5sum data2.txt**

Create a third file with different content. Note that you can copy and paste thousands of words in the third file. Calculate hash values, and note that hash length does not change depending on the input length, but the hash values change.

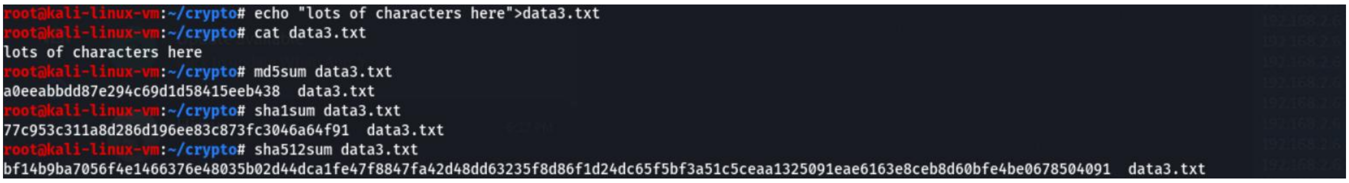
o echo "lots of characters here" > data3.txt

o cat data3.txt

• md5sum data3.txt

• sha1sum data3.txt mk

• sha512sum data3.txt



Compare the hash values of created files all at once using the md5deep tool. Note that you may need to calculate many files you collected from the suspected machine as a forensics investigator.

• md5deep data*.txt

Module Assessment

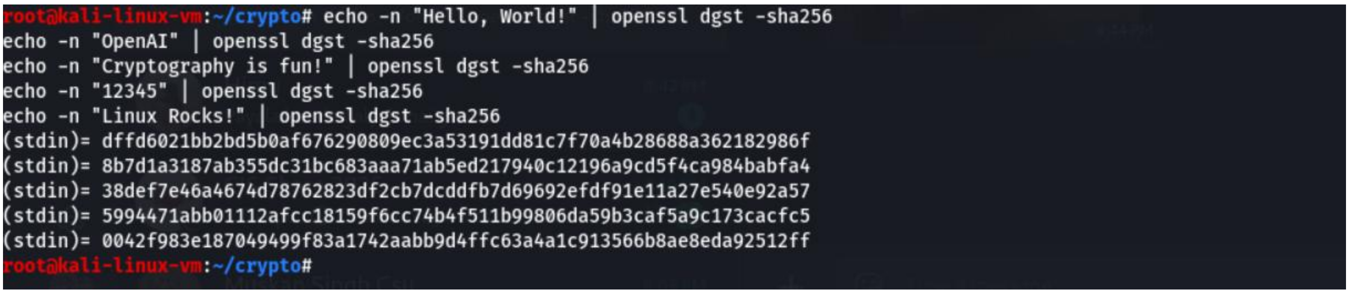
Question 1. Will the hash length change depending upon the characters of text to be encrypted ? If no, prove it with the help of screenshots .

No, the hash length does not change depending on the characters of the text being encrypted. Hash functions produce a fixed-length output regardless of the input size or content.

For example:

- **SHA-256** always produces a 256-bit hash (64 hexadecimal characters).
- **MD5** always produces a 128-bit hash (32 hexadecimal characters).

Each command will display a hash of 64 hexadecimal characters, regardless of the input text, proving that the hash length does not depend on the input content.

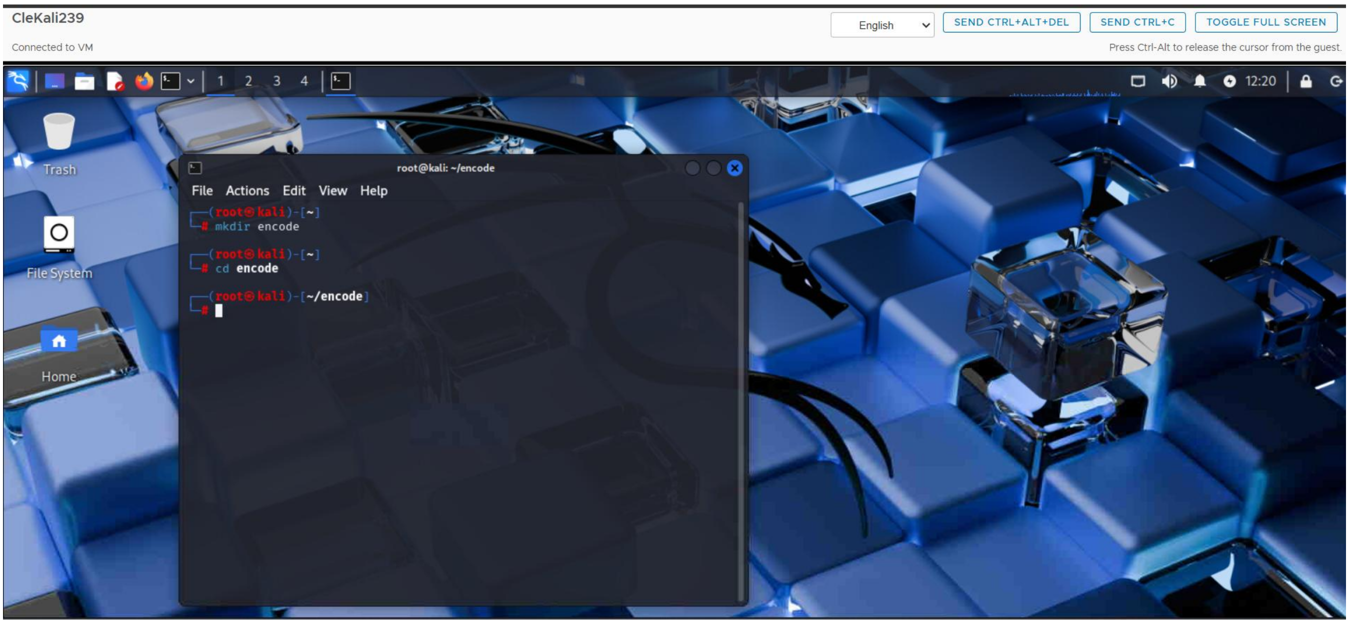


Module Activity Description

Part One: Encoding & Decoding

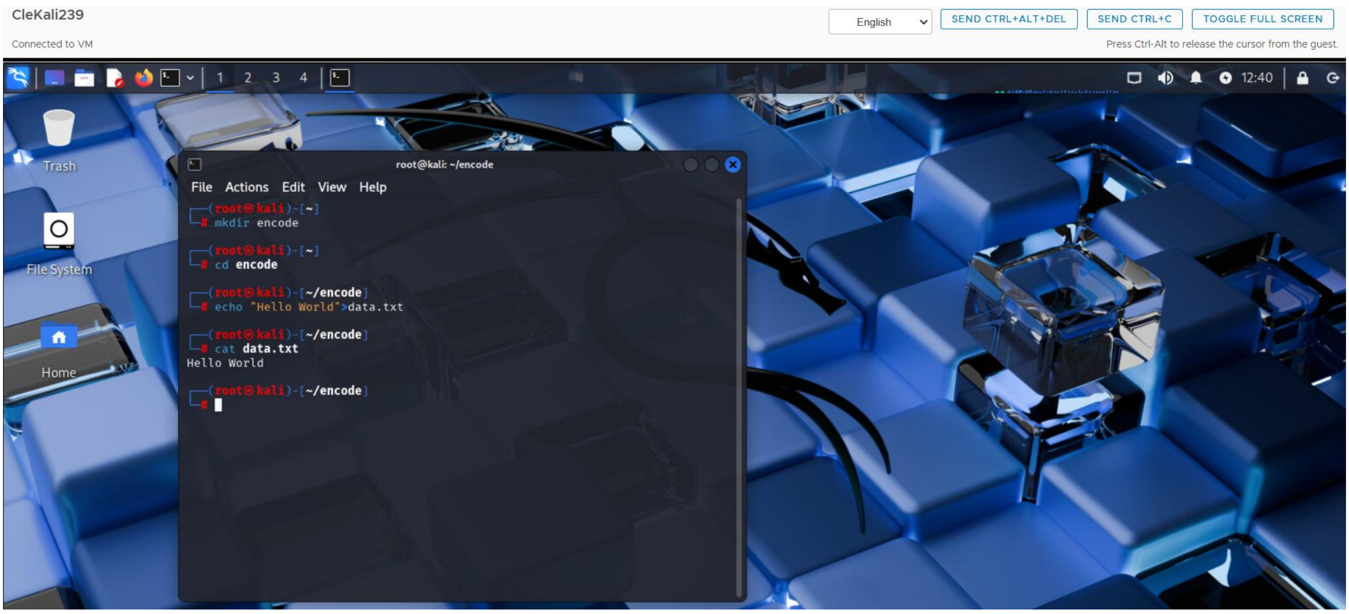
Open a terminal window on the Linux SIFT workstation. Create a folder and navigate to the newly created folder:

- **mkdir encode**
- **cd encode**



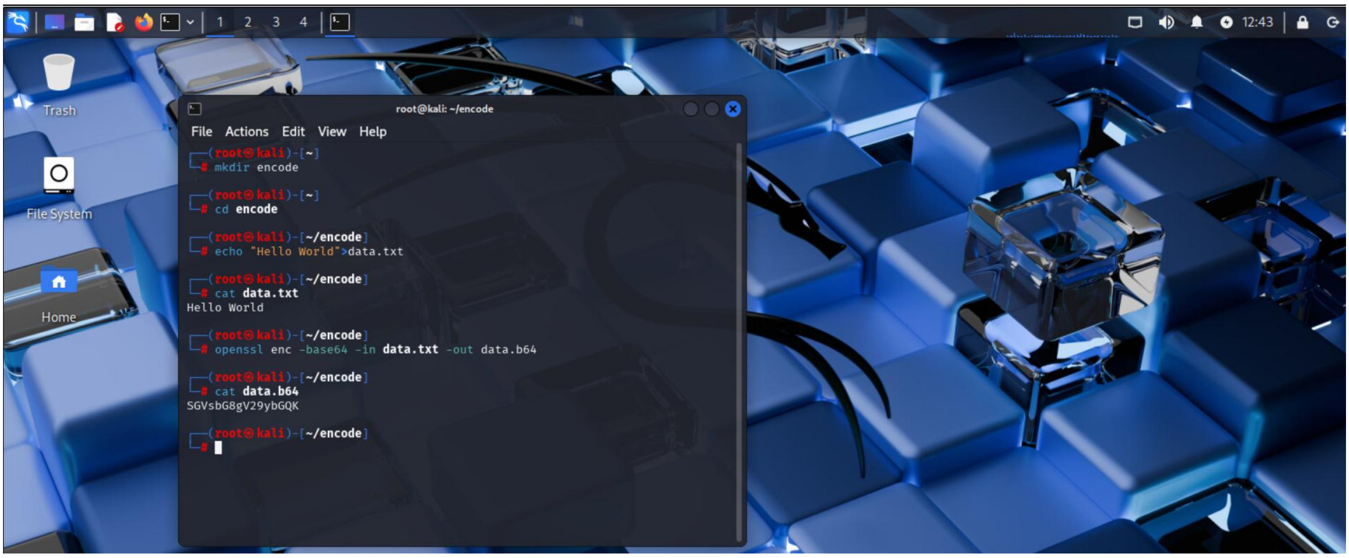
Create a file and check the content:

- echo "Hello World!" > data.txt
- cat data.txt



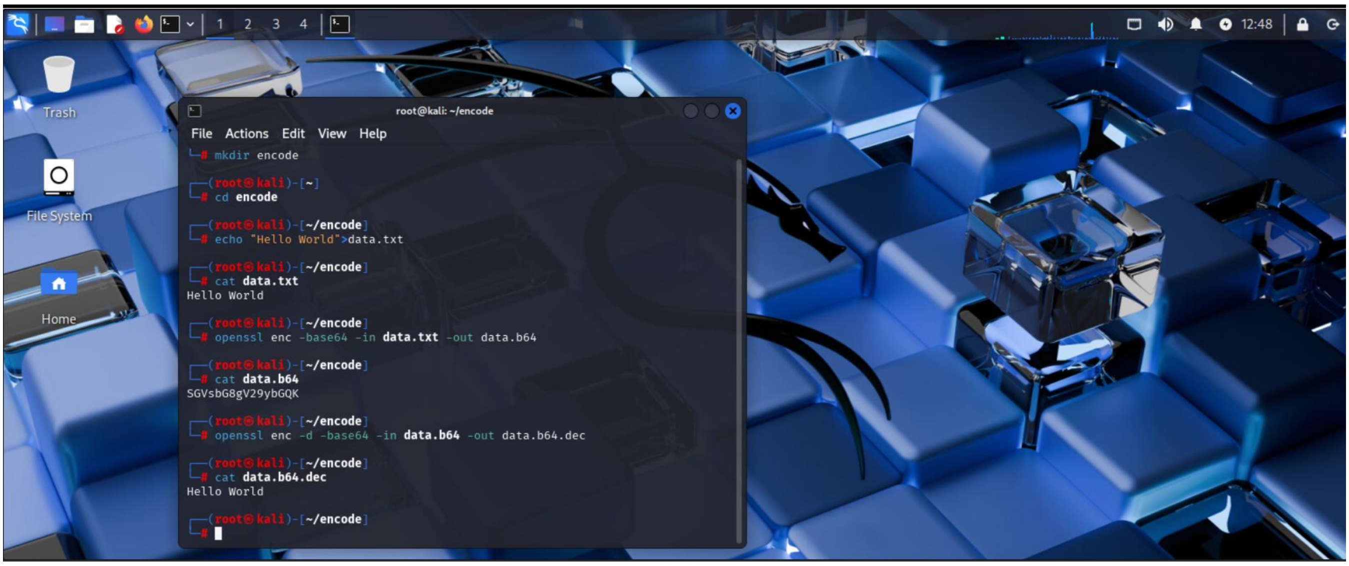
Perform Base64 encoding ofls the file using openssl utility and check the content:

- **openssl enc -base64 -in data.txt -out data.b64**
- **cat data.b64**



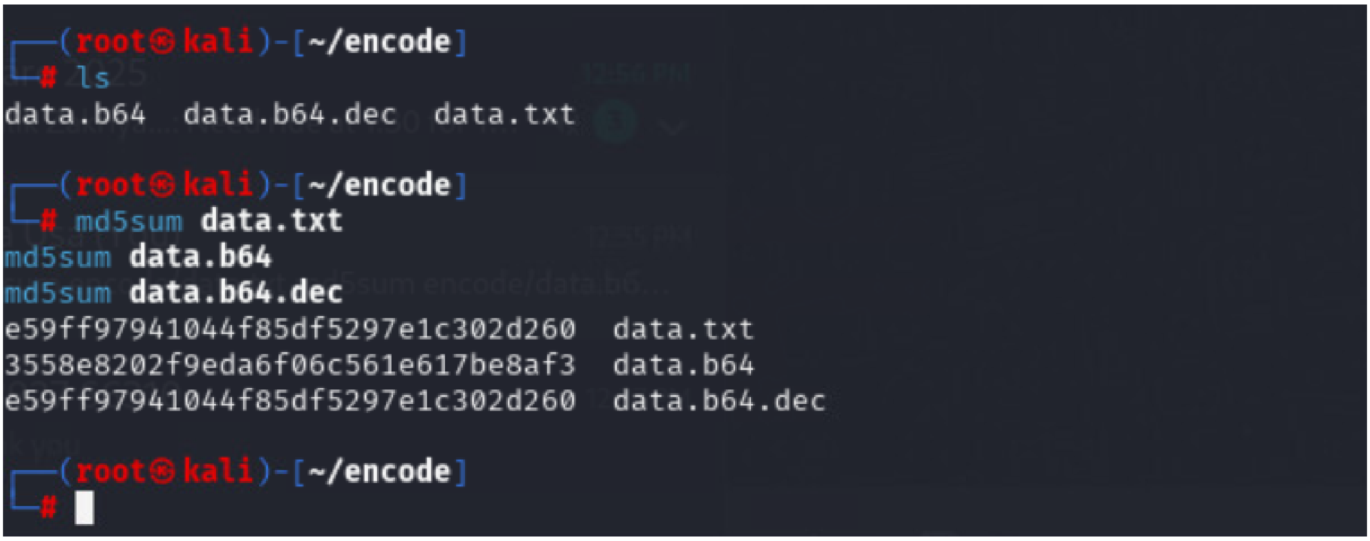
Decode the encoded file using openssl utility and check the content:

- **openssl enc -d -base64 -in data.b64 -out data.b64.dec**
- **cat data.b64.dec**



Module Assessment

Question 2. List the files from encode folder and compare the md5 hashes. What is the conclusion from comparing md5sum with the directory.



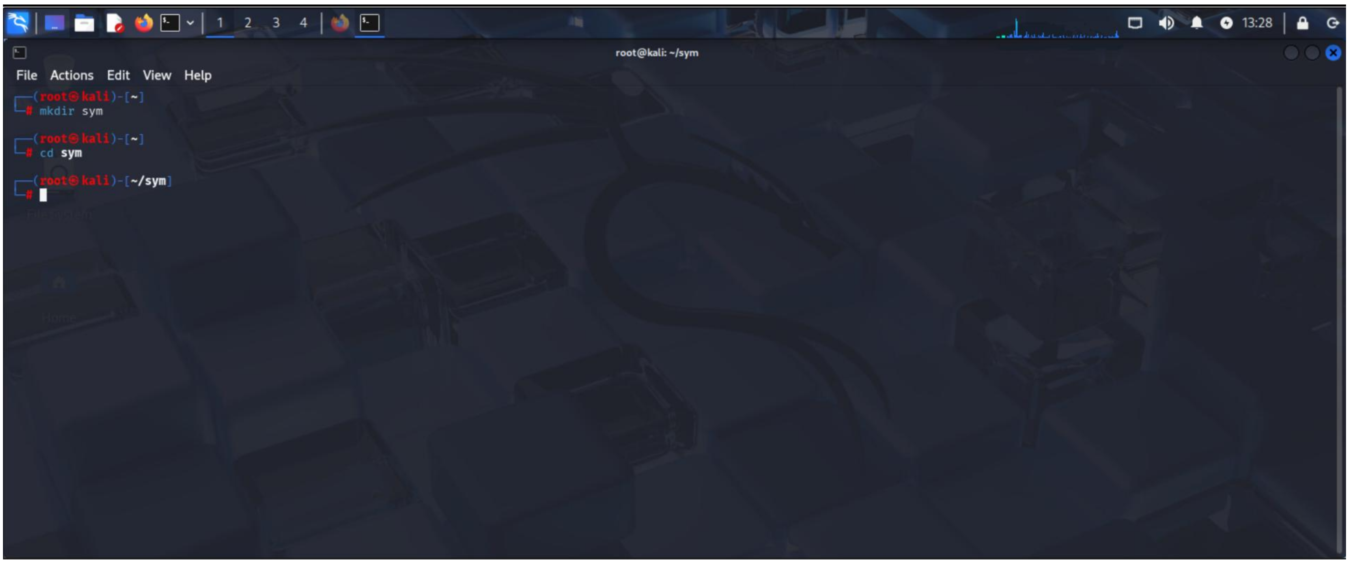
Module Activity Description:

Part Two: Symmetric Encryption

Open a terminal window on the Linux SIFT workstation. Create a folder and navigate to the newly created folder:

• mkdir sym

• cd sym



Create a file and check the content:

• echo "Hello World!" > data.txt

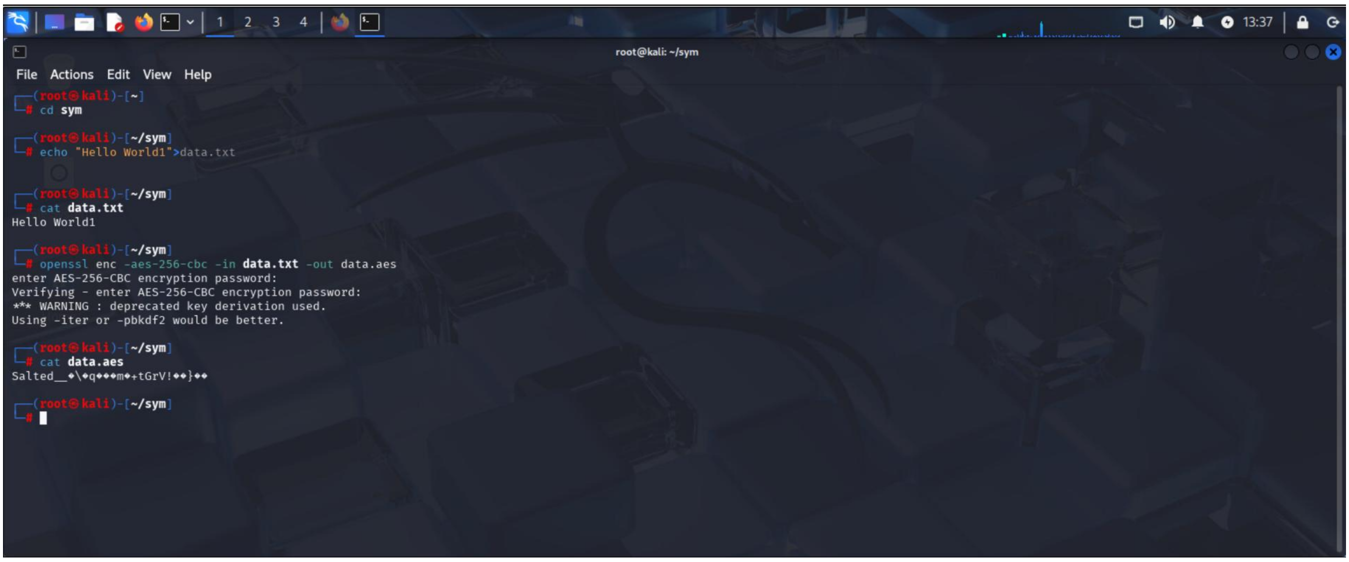
• cat data.txt

Encrypt (AES encryption) the file using openssl utility and check the content:

• openssl enc -aes-256-cbc -in data.txt -out data.aes

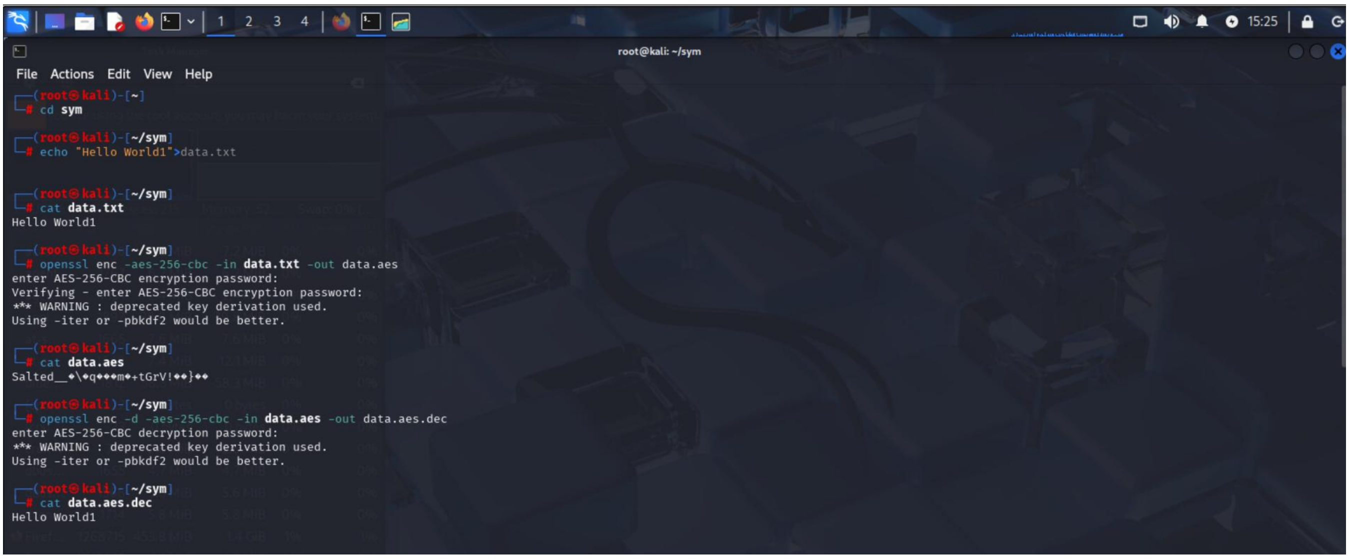
Openssl will ask for a password to protect the key. Type a password.

• cat data.aes



Decrypt the encrypted file using openssl utility and check the content:

- **openssl enc -d -aes-256-cbc -in data.aes -out data.aes.dec**
- **cat data.aes.dec**

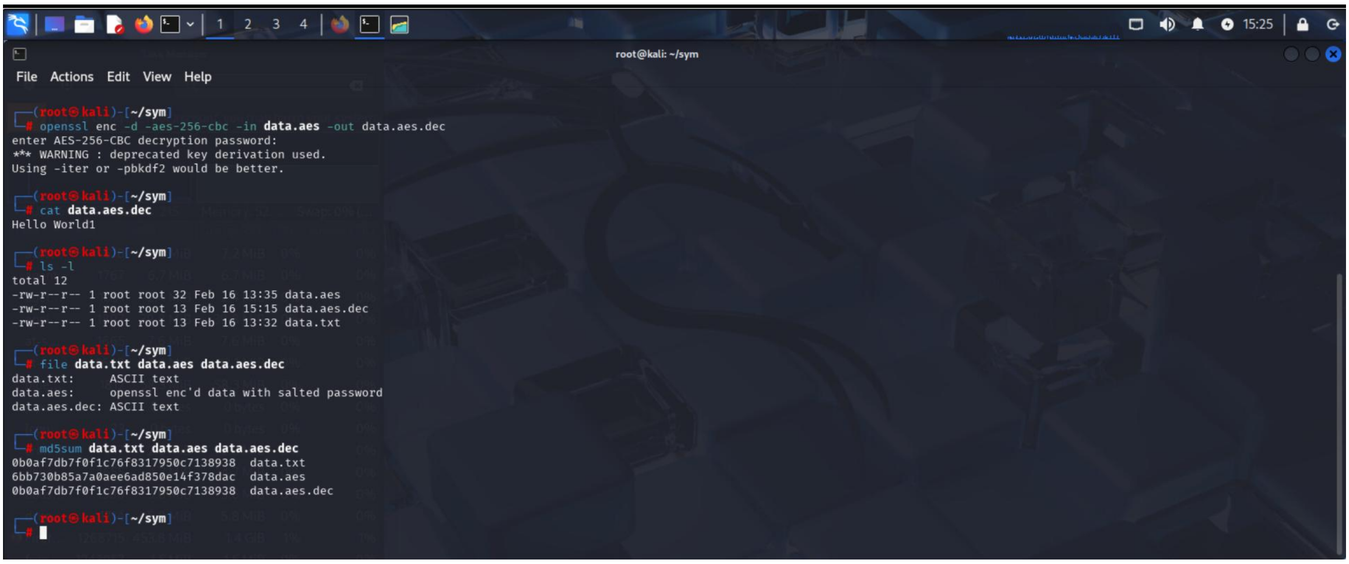


Module Assessment

Question 3 - List the files, see the filetypes, and compare the md5 hashes. What utility was used for AES encryption and explain it?

Utility: openssl

- openssl enc: Encryption command
- -aes-256-cbc: AES encryption with a 256-bit key in CBC mode
- AES (Advanced Encryption Standard) is a symmetric encryption algorithm widely used for securing data.
- CBC (Cipher Block Chaining) adds randomness by XORing each plaintext block with the previous ciphertext block before encryption.

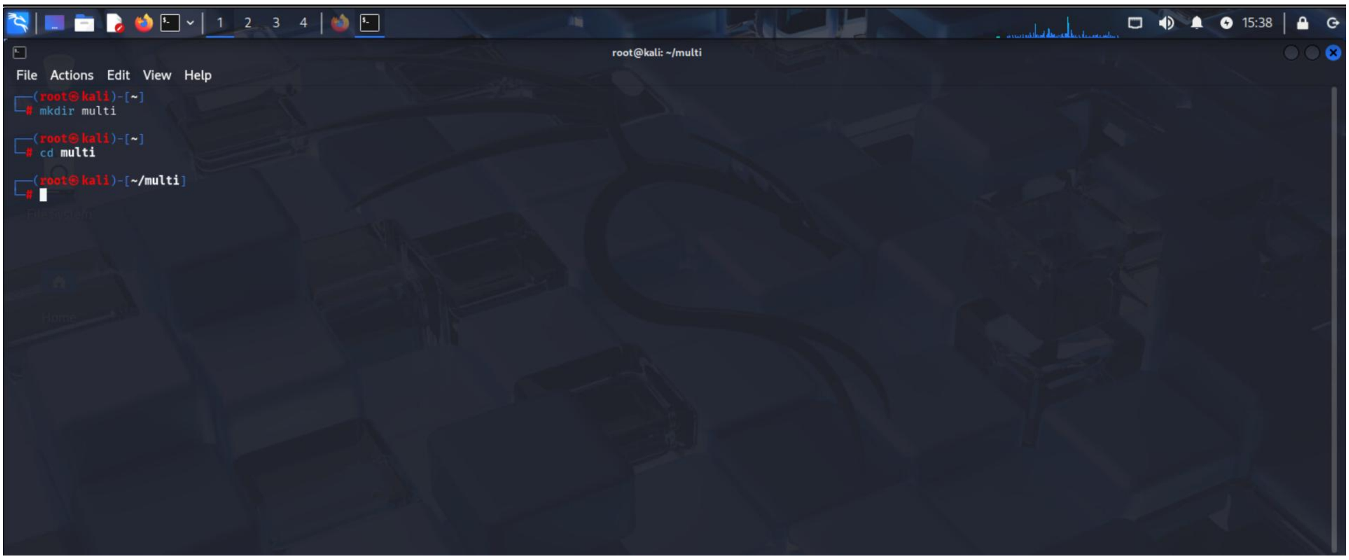


Module Activity Description:

Part Three: Multiple Operations

Open a terminal window on the Linux SIFT workstation. Create a folder and navigate to the newly created folder:

- **mkdir multi**
- **cd multi**



Create a file and check the content:

- **echo "Hello World!" > data.txt**

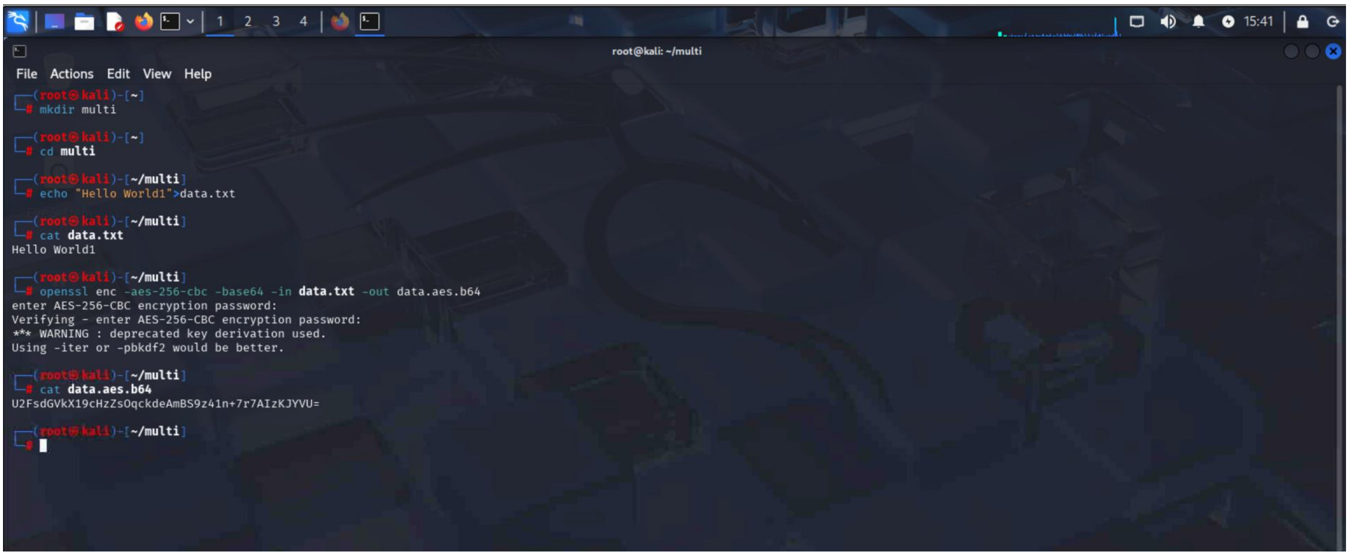
- **cat data.txt**

First encrypt the file (AES encryption) and then encode it (Base64) using openssl and check the content:

- **openssl enc -aes-256-cbc -base64 -in data.txt -out data.aes.b64**

OpenSSL will ask for a password to protect the key. Type a password.

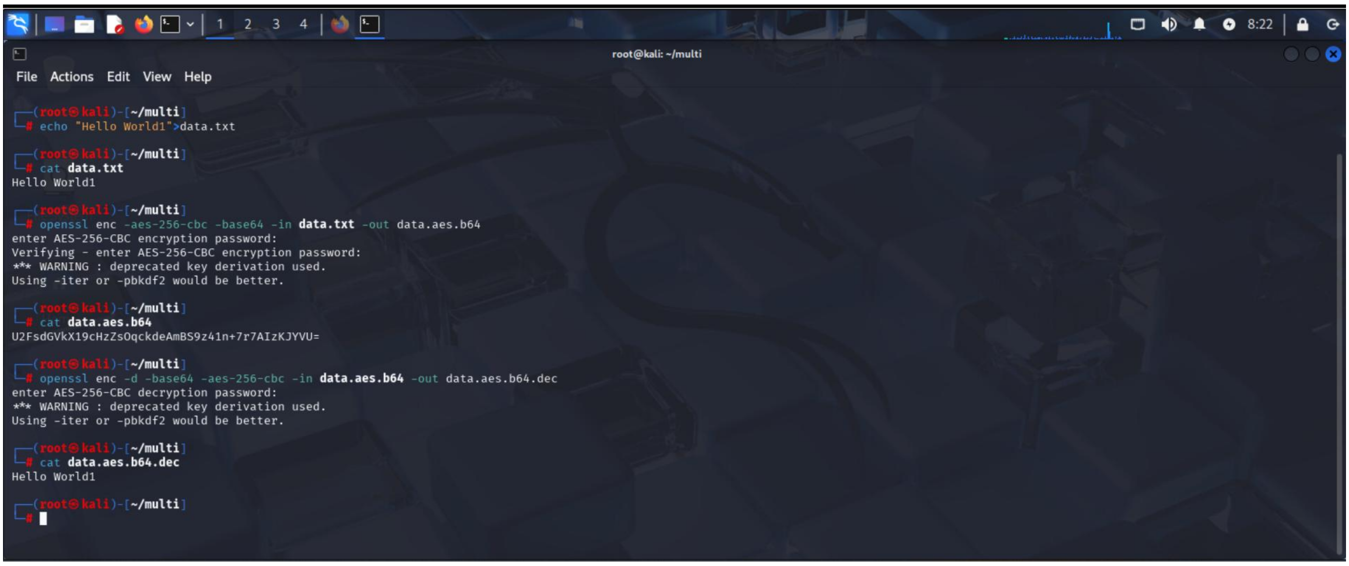
- **cat data.aes.b64**



First decode the encoded file and then decrypt the encrypted file using openssl and check the content:

- openssl enc -d -base64 -aes-256-cbc -in data.aes.b64 -out data.aes.b64.dec

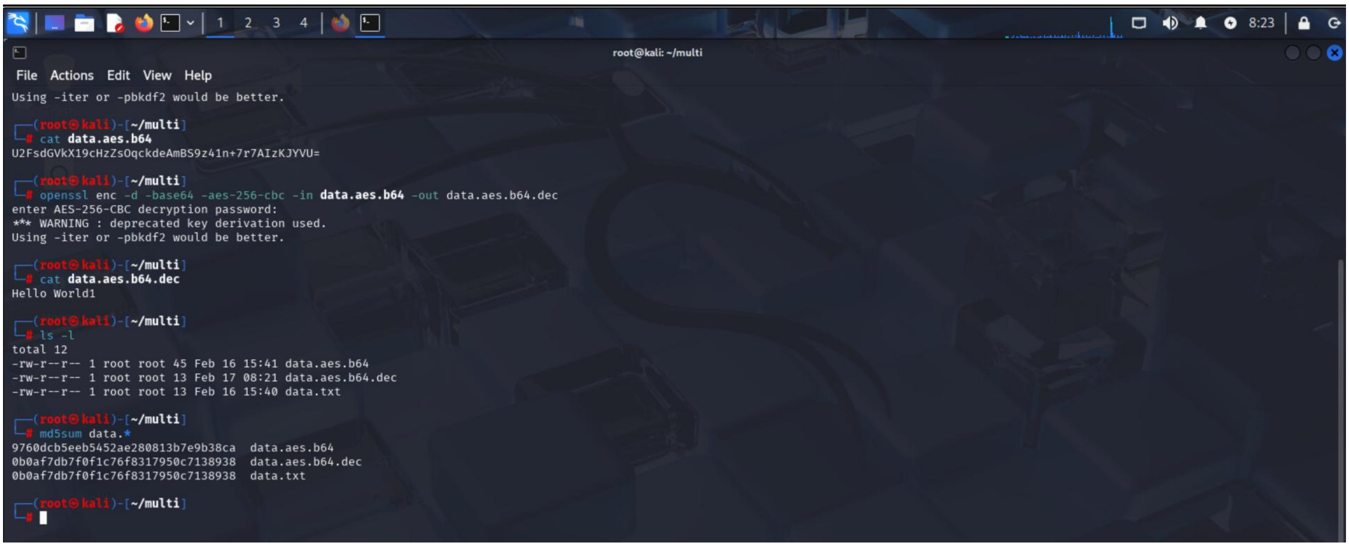
- cat data.aes.b64.dec



List the files and compare the md5 hashes:

- ls -l

- md5sum data.*



Module Assessment

Question 4 - What encoding technique was used on encryption? Compare the screenshot of encrypted file from aes encryption (part two – data.aes) and encrypted file from multiple operation (part three – data.aes.b64)? What is the difference between two files.

Explanation of Encoding Technique

The technique used was Base64 encoding after AES encryption.

- AES-256-CBC: Symmetric encryption using a 256-bit key with Cipher Block Chaining.
- Base64 Encoding: Converts binary data into ASCII characters (readable text) for easier transport and storage.

Difference between data.aes (AES only) and data.aes.b64 (AES + Base64):

- data.aes: Binary file (unreadable) with raw encrypted content.
- data.aes.b64: Text file with Base64-encoded encrypted content, making it readable (though not understandable without decryption).

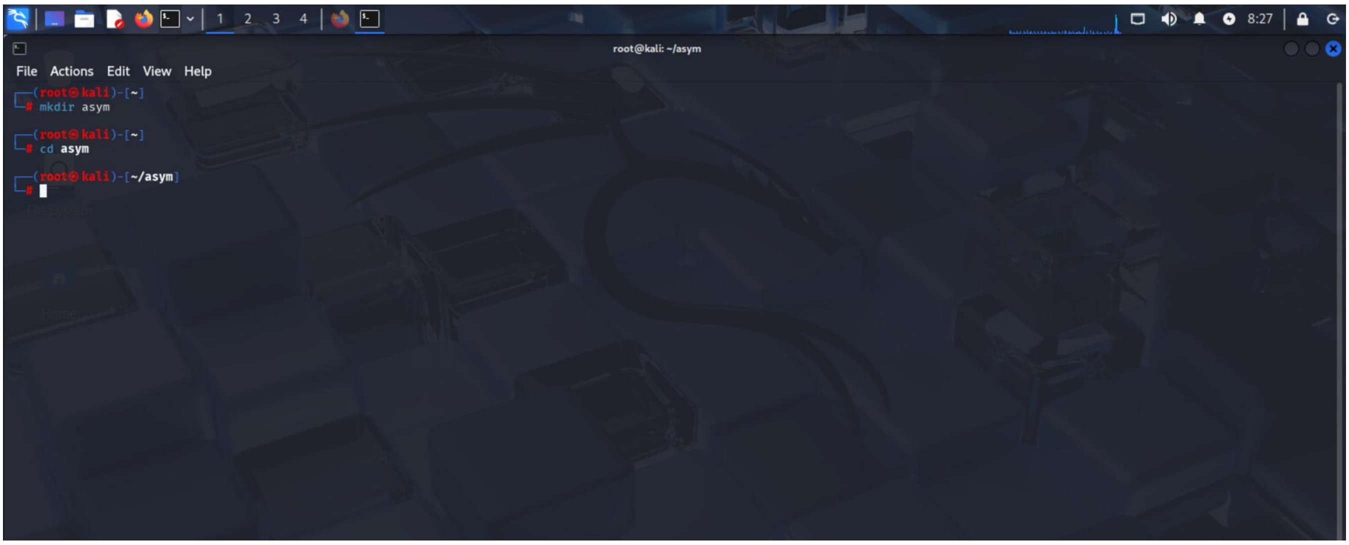
Module Activity Description:

Part Four: Asymmetric Encryption

Open a terminal window on the Linux SIFT workstation. Create a folder and navigate to the newly created folder:

- **mkdir asym**

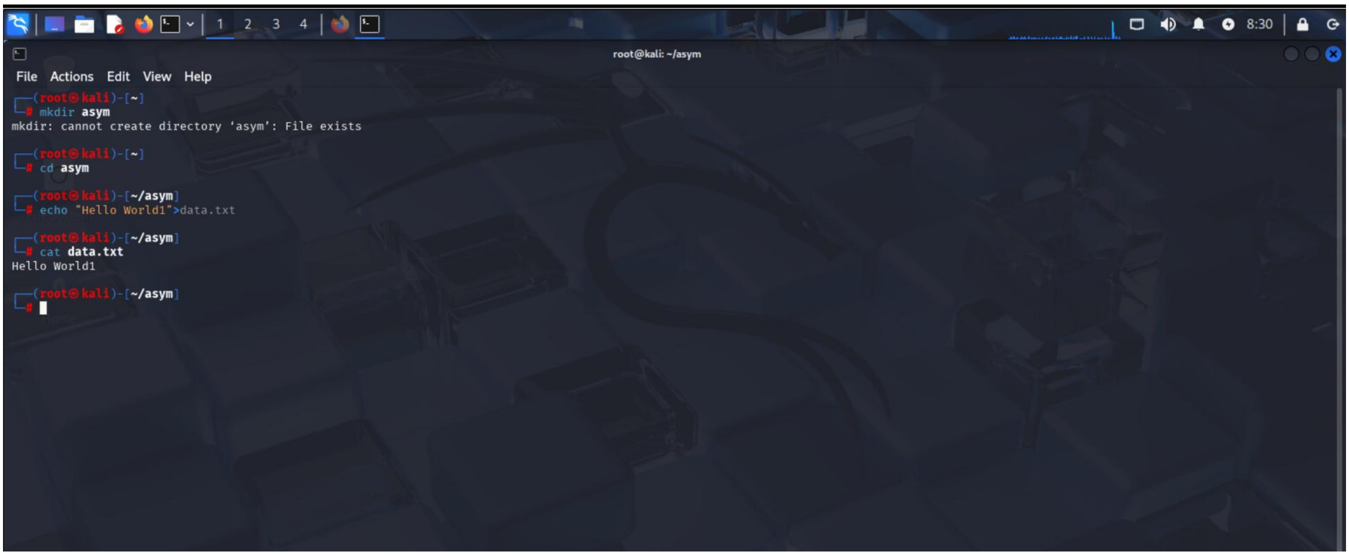
- **cd asym**



Create a file and check the content:

- **echo "Hello World! > data.txt**

- **cat data.txt**



Generate a private key using OpenSSL and check the content of the key:

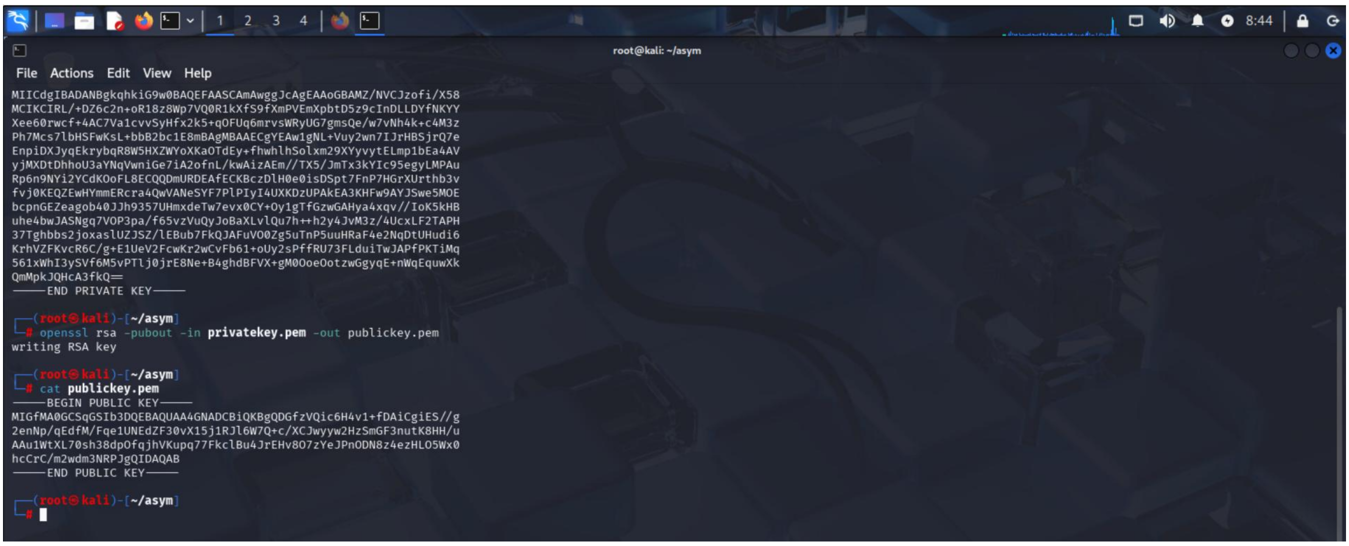
- **openssl genpkey -algorithm rsa -pkeyopt rsa_keygen_bits:1024 -out privatekey.pem**

- **cat privatekey.pem**

Generate the public key associated with the private key and check the content of the key:

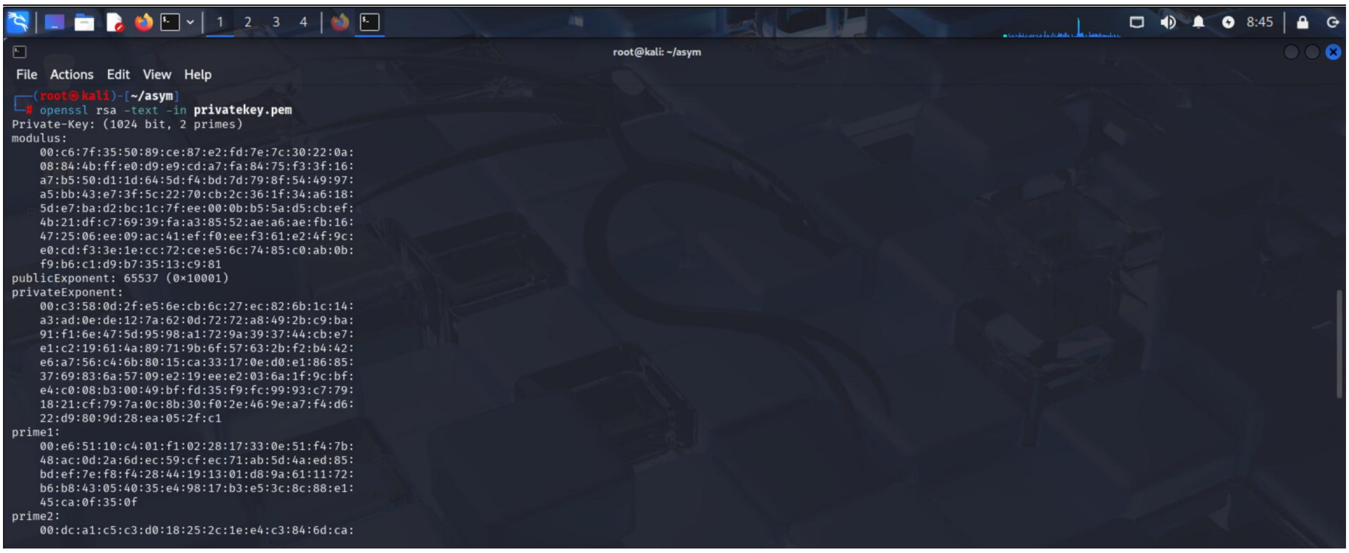
- **openssl rsa -pubout -in privatekey.pem -out publickey.pem**

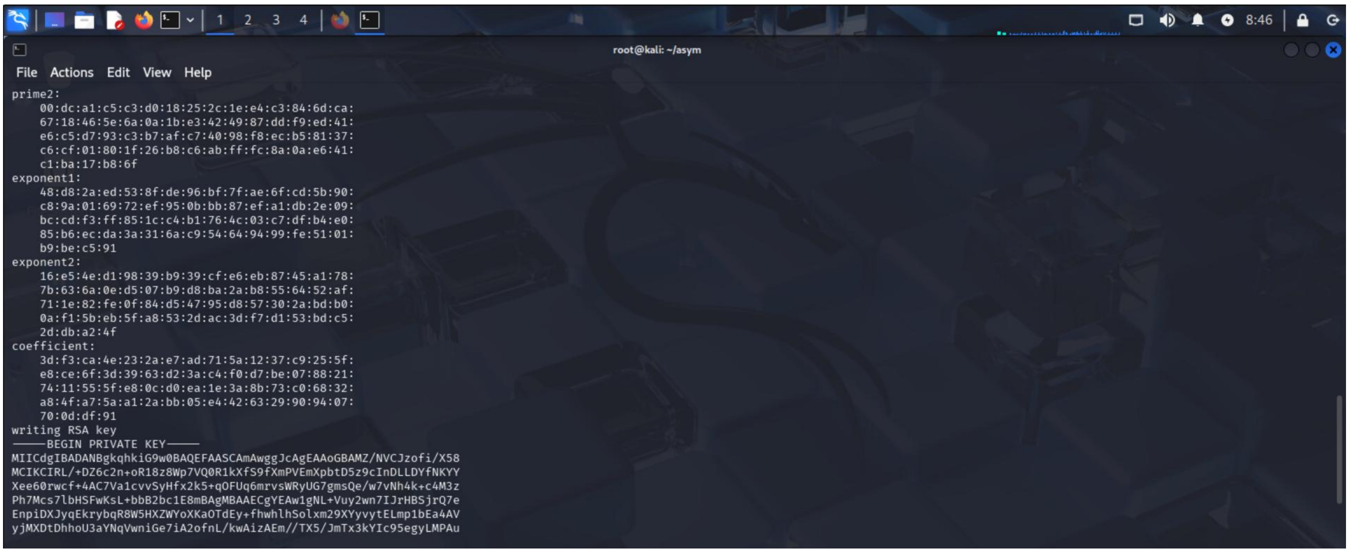
• **cat publickey.pem**



Preview the private key:

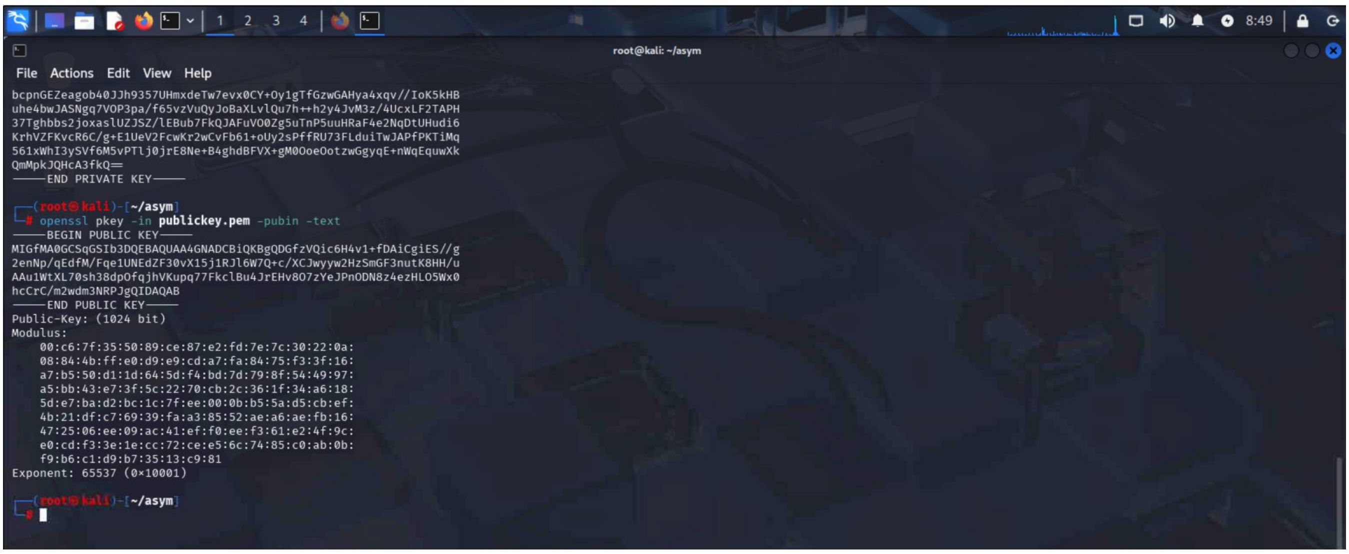
• **openssl rsa -text -in privatekey.pem**





Preview the public key:

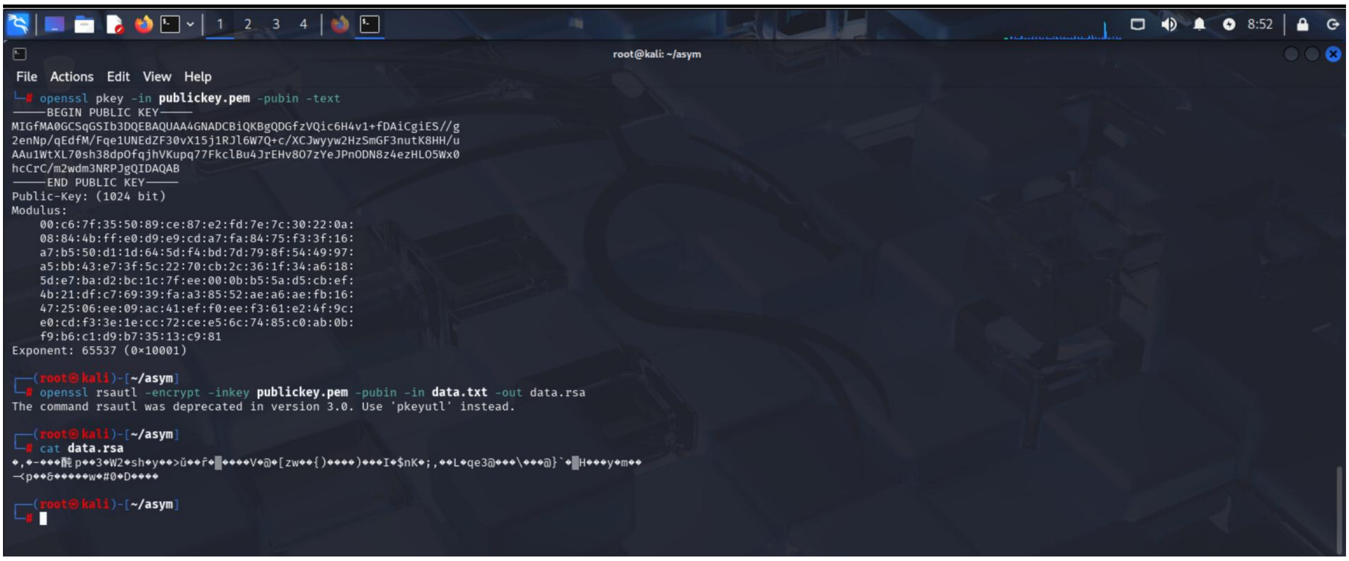
- openssl pkey -in publickey.pem -pubin -text



Encrypt the file you created and check the content:

- **openssl rsautl -encrypt -inkey publickey.pem -pubin -in data.txt -out data.rsa**

- **cat data.rsa**



Decrypt the encrypted file and check the content:

- **openssl rsautl -decrypt -inkey privatekey.pem -in data.rsa -out data.rsa.dec**

- **cat data.rsa.dec**

List the files and compare the md5 hashes:

- ls -l
- md5sum data.*

Module Assessment

Question 5 - What is a private key and what is a public key? What are the name of files used to encrypt and decrypt data.txt in this section?

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1. What is a Private Key?

- A **private key** is a confidential component of asymmetric encryption.
- It is used for **decryption** and **signing** messages.

- It must be kept secure and never shared publicly.

2. What is a Public Key?

- A **public key** is derived from the private key.
- It is used for **encryption** and **verification**.
- It can be shared openly without security risks.

Files Involved in Encryption/Decryption

Operation	File Name	Purpose
Input File	data.txt	Original plaintext file
Private Key	privatekey.pem	For decrypting data
Public Key	publickey.pem	For encrypting data
Encrypted File	data.rsa	RSA-encrypted version of data
Decrypted File	data.rsa.dec	Decrypted plaintext file

Module Activity Description:

Part Five: Digital Signatures

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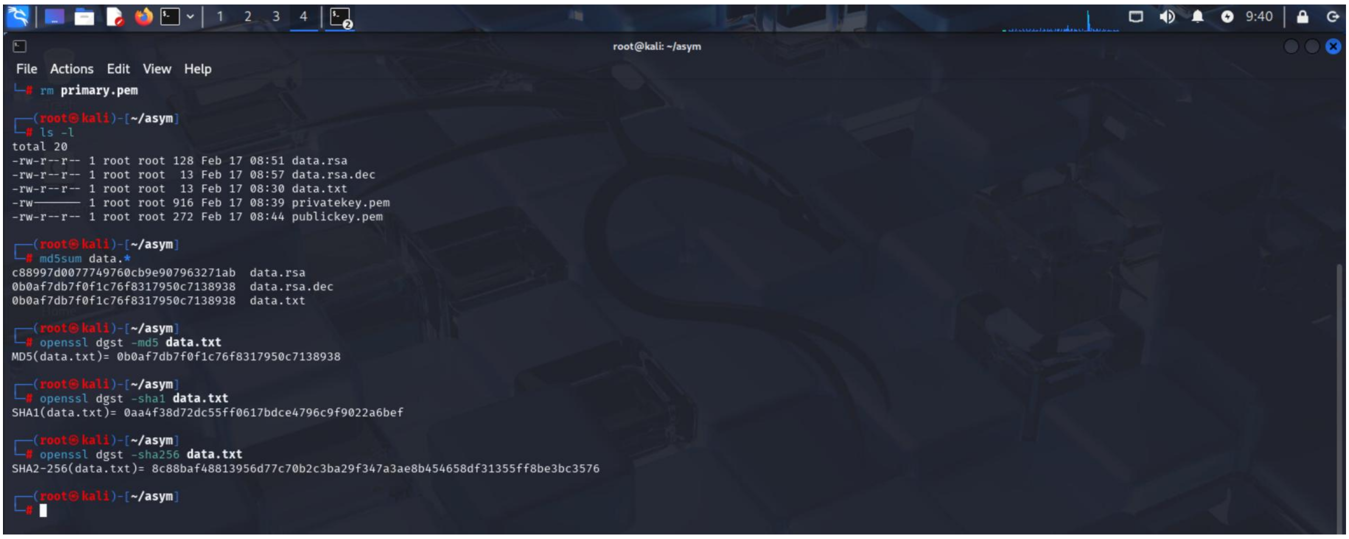
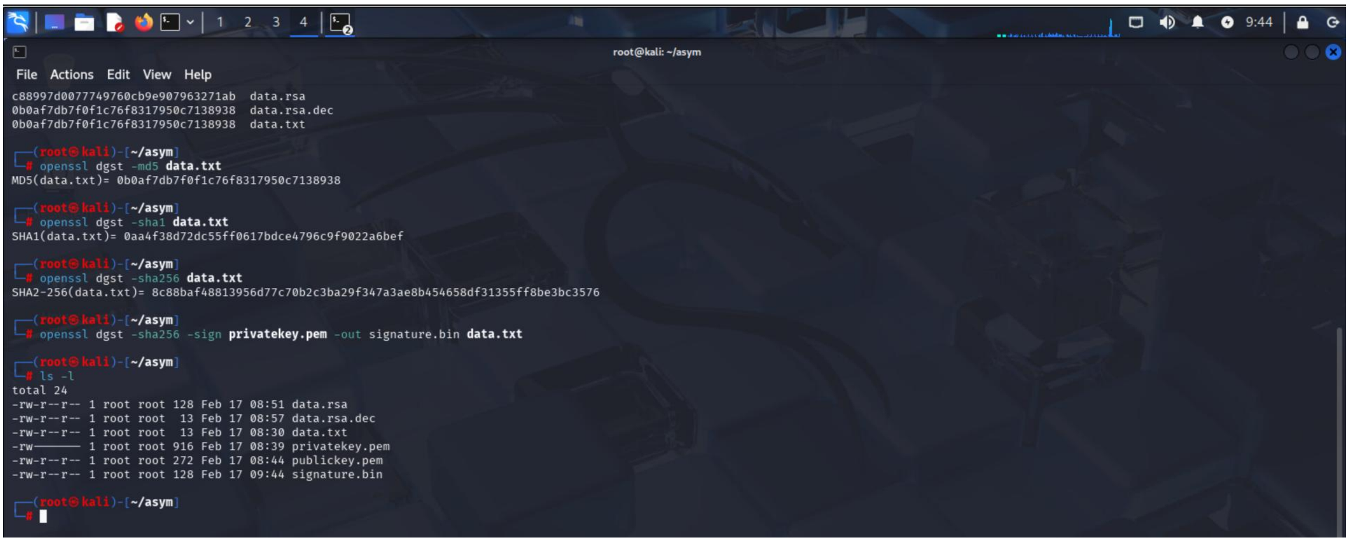
1. **Paste the screenshot of different hash values of the data file (data.txt).**

   - **openssl dgst -md5 data.txt**
   - **openssl dgst -sha1 data.txt**
   - **openssl dgst -sha256 data.txt**



Sign the file and check the content of the signature:

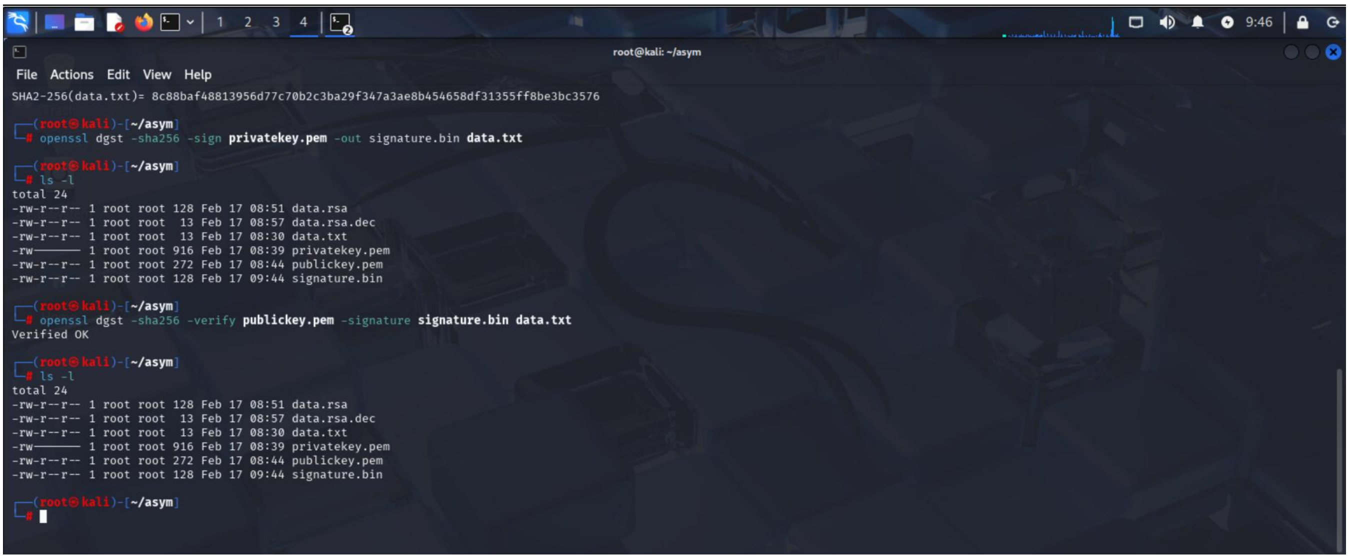
- **openssl dgst -sha256 -sign privatekey.pem -out signature.bin data.txt**
- **ls -l**



Receiver to verify the signature:

- **openssl dgst -sha256 -verify publickey.pem -signature signature.bin data.txt**

- **ls -l**



Module Assessment

Question 6 - What was used to verify the signature of data.txt. Will verification fail if any of the file – signature.bin, publickey.pem, data.txt changes?

What was used to verify the signature?

- The public key (publickey.pem) was used to verify the signature.
- OpenSSL calculates the SHA256 hash of data.txt, decrypts signature.bin with the public key, and compares the two hashes.
- If the hashes match, verification succeeds.

Will verification fail if any file changes?

Yes, verification will fail if any of the following changes:

1. data.txt: Changes in content will alter the hash and cause verification failure.
2. signature.bin: If tampered with, it will not correctly match the file's hash.
3. publickey.pem: If replaced with a different key, the signature will not decrypt correctly.